OPTIMAL UPPER BOUND OF VOLUMES FOR VARIETIES WITH LARGE SESHADRI CONSTANTS

JIHAO LIU

Abstract. We prove that if the anti-canonical divisor of a normal projective variety has moving Seshadri constant greater than the dimension minus one, then the product of the anti-canonical volume and the excess of the moving Seshadri constant over the dimension minus one is at most twice the anti-canonical volume of the projective space of equal dimension, with equality for the projective space. In particular, the anti-canonical volume is at most twice Euler's number times the dimension to the power of the dimension, divided by the excess of the moving Seshadri constant over the dimension minus one, and the constant twice Euler's number is optimal. We thus answer Question 8.1 of Zhuang's paper on Fano varieties with large Seshadri constants in the affirmative.

Contents

1. Introduction 1
2. Preliminaries 5
3. Reduction to Mori fiber spaces 7
4. Fibrations over the projective line 9
5. Fano varieties of Picard number one 15
6. Proofs of the main theorems 17
References 20

1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Let L be an ample line bundle on a projective manifold X. By [Dem92, Theorem 6.4], the Seshadri constant of L at any point $x \in X$ equals $\lim_{m\to+\infty} s(mL,x)/m$, where $s(mL,x)$ is the largest integer s such that $H^0(X,L^{\otimes m})$ generates s-jets at x. If L is only nef, [Dem92, Theorem 6.4] shows that the Seshadri constant of L at x is at least $\limsup_{m\to+\infty} s(mL,x)/m$, with equality off a divisor if L is moreover big. Zhuang [Zhu17, formula (1) and the paragraph following it] takes the asymptotic order of jets as the definition of the moving Seshadri constant $\varepsilon_{\mathrm{mov}}(D,x)$ of a $\mathbb{Q}$-divisor D on a normal projective variety X at a smooth point x, and defines $\varepsilon_{\mathrm{mov}}(D)$ as the maximum of $\varepsilon_{\mathrm{mov}}(D,x)$ over smooth points x. The terminology goes back to [Nak03], and the jet definition coincides with the definition in [Nak03] when D is $\mathbb{Q}$-Cartier [ELM+09, Proposition 6.6]. In particular, $\varepsilon_{\mathrm{mov}}(-K_X)$ and the volume $\mathrm{vol}(-K_X)$, which are defined through the reflexive sheaves $\mathcal{O}_X(-mK_X)$ for all positive integers m (Definition 2.2), need no assumption on the singularities of X.

The moving Seshadri constant behaves well under the minimal model program: by [Zhu17, Corollary 3.3], $\varepsilon_{\mathrm{mov}}(-K_X) \le \varepsilon_{\mathrm{mov}}(-K_Y)$ for any birational map $X \dashrightarrow Y$ of normal projective varieties whose inverse contracts no divisor, in particular for divisorial contractions and flips. Zhuang proved that a normal projective variety X of dimension d with $\varepsilon_{\mathrm{mov}}(-K_X) > d$ is

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isomorphic to $\mathbb{P}^d$ [Zhu17, Theorem 1.7]. Earlier, $X \cong \mathbb{P}^d$ was proved under the assumption that the Seshadri constant of $-K_X$ is larger than d, for Fano manifolds [BS09, Theorem 1.7] and for $\mathbb{Q}$-Fano varieties [LZ18, Theorem 2], and in positive characteristic under the assumption that it is at least $d+1$ [Mur18].

When $d-1 < \varepsilon_{\mathrm{mov}}(-K_X) \leq d$, the variety X is no longer determined by $\varepsilon_{\mathrm{mov}}(-K_X)$, but the anti-canonical volume is bounded. For any positive real number ϵ, Zhuang proved that $\mathrm{vol}(-K_X)$ is bounded from above by a constant depending only on d and ϵ for any normal projective variety X of dimension d with $\varepsilon_{\mathrm{mov}}(-K_X) > d-1+\epsilon$ [Zhu17, Theorem 3.6], and observed that the constant can be taken of order $O(d^{2d}/\epsilon^d)$ [Zhu17, Section 4].

Zhuang's argument in [Zhu17, Section 4] only yields the order ϵ^{-d}. He argues by contradiction using the connectedness of non-klt centers of a pair whose anti-log canonical divisor is ample. The volume enters only through two $\mathbb{Q}$-divisors $D' \sim_{\mathbb{Q}} -bK_X$ and $D \sim_{\mathbb{Q}} -cK_X$ given by a dimension count, where D' has multiplicity less than 1 at a very general point x and D has multiplicity at least d at another point. The dimension count producing D needs $c^d \,\mathrm{vol}(-K_X) > d^d$. On the other hand, the part of the boundary used for the jet condition at x is $\mathbb{Q}$-linearly equivalent to $-aK_X$ and has multiplicity $a(d-1+\epsilon)$ at x, which must be larger than $d-1$ since D' has multiplicity less than 1 at x and the total multiplicity at x must be at least d. Since the anti-log canonical divisor of the pair is ample, $a+b+c<1$, which leaves only $b+c<1-a<\epsilon/(d-1+\epsilon)$ for D' and D. Thus the contradiction is reached only when

$$\mathrm{vol}(-K_X) > \Big(\frac{d(d-1+\epsilon)}{\epsilon}\Big)^d,$$

which is of order d^{2d}/ϵ^d for $d \geq 2$ and $\epsilon \leq 1$. In other words, the volume enters through its d-th root, and the coefficient available for D is of order ϵ/d.

Moreover, the assumption $\epsilon > 0$ is necessary, and the order ϵ^{-1} cannot be improved. Indeed, assume that $d \geq 2$, and for an integer $a \geq 2$ let $X_a := \mathbb{P}(1,1,a,\ldots,a)$, with $d-1$ weights equal to a. Then $-K_{X_a}$ is an ample $\mathbb{Q}$-Cartier divisor, and by [LZ18, Example 22], [Zhu17, Example 1.3] (the point $[1:0:\cdots:0]$ has an open $\mathrm{Aut}(X_a)$-orbit), and Lemma 6.1,

$$\varepsilon_{\mathrm{mov}}(-K_{X_a}) = d-1+\frac{2}{a} \quad \text{and} \quad \mathrm{vol}(-K_{X_a}) = \frac{(2+(d-1)a)^d}{a^{d-1}}.$$

Thus $(\varepsilon_{\mathrm{mov}}(-K_{X_a})-d+1)\,\mathrm{vol}(-K_{X_a}) = 2(d-1+2/a)^d$, while $\mathrm{vol}(-K_{X_a}) \to +\infty$ as $a \to +\infty$ (hence for $0<\epsilon<1$ and $a := \lceil 2/\epsilon \rceil - 1 \geq 2$, we have $\varepsilon_{\mathrm{mov}}(-K_{X_a}) > d-1+\epsilon$ and $\mathrm{vol}(-K_{X_a}) \geq a(d-1)^d \geq (2/\epsilon-1)(d-1)^d$).

The varieties X_a also show that the volume cannot be bounded on a birational model. Zhuang proved that $\varepsilon_{\mathrm{mov}}(-K_X) > d-1$ implies that X is birational to a terminal Fano variety Y with $\varepsilon_{\mathrm{mov}}(-K_Y) \geq \varepsilon_{\mathrm{mov}}(-K_X)$ [Zhu17, Theorem 3.7]. In [Zhu17, Theorem 3.7] the left-hand side is the Seshadri constant of $-K_Y$, which is equal to $\varepsilon_{\mathrm{mov}}(-K_Y)$ by Lemma 6.1 since $-K_Y$ is ample and $\mathbb{Q}$-Cartier. Together with [Bir16, Theorem 1.1], such varieties X form a birationally bounded family. Zhuang runs an MMP [Zhu17, Section 5]. When the output is a Mori fiber space over $\mathbb{P}^1$, he replaces it by $\mathbb{P}^d$ using rationality. However, the varieties X_a are toric, hence rational, and $\mathrm{vol}(-K_{X_a}) \to +\infty$, whereas $\mathrm{vol}(-K_{\mathbb{P}^d}) = (d+1)^d$. Thus, to bound the volume, one cannot replace X by a birational Fano model, and the Mori fiber spaces over $\mathbb{P}^1$ have to be treated directly. Zhuang asked for the optimal order of the bound on $\mathrm{vol}(-K_X)$ in [Zhu17, Theorem 3.6].

Question 1.1 ([Zhu17, Question 8.1])**.** For a positive integer d and a positive real number ϵ, let $M(d,\epsilon)$ be the supremum of $\mathrm{vol}(-K_X)$, where X runs over all normal projective varieties of dimension d such that $\varepsilon_{\mathrm{mov}}(-K_X) > d-1+\epsilon$. Is $M(d,\epsilon) = O(d^d/\epsilon)$?

If the implied constant may depend on d, the question asks whether $M(d,\epsilon) \leq C_d/\epsilon$. Otherwise it asks whether $M(d,\epsilon) \leq Cd^d/\epsilon$ for an absolute constant C. Before this paper, the following cases of Question 1.1 and a variant in positive characteristic were known.

(1) If $\epsilon \geq 1$, then $X \cong \mathbb{P}^d$ by [Zhu17, Theorem 1.7], so $M(d,\epsilon) \leq (d+1)^d$.

(2) In positive characteristic and dimension three, for any $\epsilon > 0$ there exists a constant depending only on ϵ that bounds $\mathrm{vol}(-K_X)$ from above for any normal projective threefold X over an algebraically closed field of characteristic $p > 5$ with $\varepsilon_{\mathrm{mov}}(-K_X, x) > 2 + \epsilon$ for some smooth point $x \in X$ [Zhu22, Theorem 1.9].

(3) If $d = 2$, Zhuang sketched a proof of $M(2, \epsilon) = O(\epsilon^{-1})$, without an explicit constant [Zhu17, Section 8, the paragraph after Question 8.1].

The anti-canonical volume and the moving Seshadri constant do not decrease under the minimal resolution and under contracting (-1)-curves, so it suffices to examine ruled surfaces. Indeed, a terminal surface is smooth [KM98, Theorem 4.5(1)], and a smooth projective surface S with a K_S-negative extremal contraction to a point has $\rho(S) = 1$ and $-K_S$ ample [KM98, Theorem 1.28(3)]. Then $H^1(S, \mathcal{O}_S) = H^2(S, \mathcal{O}_S) = 0$ by [KM98, Theorem 2.47] and Serre duality, and $H^0(S, \mathcal{O}_S(2K_S)) = 0$ since $-K_S$ is ample, so S is rational [Har77, Theorem V.6.2]. Since $\rho(S) = 1$, S contains no (-1)-curve, hence is a relatively minimal model [Har77, Corollary V.5.4], and S is not one of the rational ruled surfaces, which have Picard number 2 [Har77, Proposition V.2.3]. Thus $S \cong \mathbb{P}^2$ [Har77, Example V.5.8.2 and Remark V.5.8.4]. We have $\varepsilon_{\mathrm{mov}}(-K_{\mathbb{P}^2}) = 3$ by [Dem92, Theorem 6.4] and [Laz04, Example 5.1.7], and $\mathrm{vol}(-K_{\mathbb{P}^2}) = 9$ by Lemma 6.1, so $\mathrm{vol}(-K_X) \leq 9$ when the output is $\mathbb{P}^2$.

For $d \geq 3$, the MMP of [Zhu17, Section 5] may end with a terminal Fano variety Y of Picard number one with $Y \ncong \mathbb{P}^d$, hence $\varepsilon_{\mathrm{mov}}(-K_Y) \leq d$ by [Zhu17, Theorem 1.7]. For example, a smooth quadric hypersurface $Q \subset \mathbb{P}^{d+1}$ satisfies $\rho(Q) = 1$ [Har77, Exercise II.6.5(c)] and $-K_Q \sim dH$ for a hyperplane section H. Since Q is covered by lines, $\varepsilon_{\mathrm{mov}}(-K_Q) = d$ [Dem92, Theorem 6.4], [Laz04, Example 5.1.6], while $((-K_Q)^d) = 2d^d$. So, unlike for surfaces, an output over a point need not be $\mathbb{P}^d$, and its volume has to be bounded without listing such varieties. Moreover, if the output is a Mori fiber space $f : Y \to Z$ with $\dim Z > 0$, then $Z \cong \mathbb{P}^1$ and the general fiber of f is $\mathbb{P}^{d-1}$ [Zhu17, Lemma 5.3], but such fibrations cannot be listed and computed one by one as ruled surfaces. To our knowledge, no bound of order ϵ^{-1} was known for $d \geq 3$, even with a constant depending on d.

Known volume bounds for Fano varieties do not answer Question 1.1. By [Bir16, Theorem 1.1], terminal Fano varieties of dimension d form a bounded family, so their anti-canonical volumes are bounded by a constant depending only on d, but the dependence on d is not explicit. The sharp bound $((-K_X)^d) \leq (d+1)^d$ holds for Ding-semistable $\mathbb{Q}$-Fano varieties [Fuj18, Corollary 1.3], see also [Liu18, Theorem 16] applied to a reduced smooth point of X, hence for K-semistable ones, since Ding-semistability and K-semistability are equivalent for $\mathbb{Q}$-Fano varieties (see [Fuj18, Remark 1.4]), and a K-semistable Fano manifold X of dimension d with $X \ncong \mathbb{P}^d$ satisfies $((-K_X)^d) \leq 2d^d$ [LM25, Theorem 1.1]. Zhang conjectured that any K-semistable Fano manifold X of dimension d satisfies $h^0(X, -mK_X) \leq h^0(\mathbb{P}^d, -mK_{\mathbb{P}^d})$ for any $m \geq 1$, with equality for some m only if $X \cong \mathbb{P}^d$ [Zha25, Conjecture 1.2]. [LZ26, Theorem 2.1] proves the inequality, with equality for some m only if $X \cong \mathbb{P}^d$, when T_X is semistable with respect to $-K_X$, by counting jets along the filtration by principal parts [LZ26, Lemma 2.3], and this implies Zhang's conjecture [LZ26, Corollary 2.7].

[BY26, Theorem 3.2] extends the count to projective klt varieties and removes the semistability of T_X, keeping all slopes of the Harder–Narasimhan filtration of T_X. For a klt Fano variety X of dimension d, let $\lambda_1, \ldots, \lambda_d$ be the slopes of the Harder–Narasimhan filtration of T_X with respect to $-K_X$, repeated according to rank and divided by $((-K_X)^d)$, so that $\sum_{i=1}^d \lambda_i = 1$. If $\lambda_1, \ldots, \lambda_d$ are all positive, then

$$((-K_X)^d) \leq \left(\frac{d+1}{d}\right)^d \frac{1}{\prod_{i=1}^d \lambda_i}$$

[BY26, Theorem 3.2 and Remark 3.3]. Section 5 uses the inequality. The bound is weak when $\prod_{i=1}^d \lambda_i$ is small. The birational boundedness of Fano type fibrations [Jia18, Theorem 1.4], which

is conditional on conjectures in lower dimensions, and their boundedness [Bir22, Theorem 1.2] assume that the base belongs to a bounded family and that the fibration has bounded degree with respect to a polarization of the base. Such bounds are not expressed through $\varepsilon_{\mathrm{mov}}(-K_X)$. None of K-semistability, semistability of T_X, positivity of $\lambda_1, \dots, \lambda_d$, or bounds on fibrations is part of the hypotheses of Question 1.1.

The goal of this paper is to answer Question 1.1.

We answer Question 1.1 positively and determine the optimal absolute constant. The following theorem and corollary are proved in Section 6.

Theorem 1.2. *Let d be a positive integer and X a normal projective variety of dimension d. Assume that $\varepsilon_{\mathrm{mov}}(-K_X) > d-1$. Then*

$$(\varepsilon_{\mathrm{mov}}(-K_X) - d + 1)\,\mathrm{vol}(-K_X) \leq 2(d+1)^d.$$

Moreover, the equality holds if $X \cong \mathbb{P}^d$.

Corollary 1.3. *Let d be a positive integer, ϵ a positive real number, and X a normal projective variety of dimension d. Assume that $\varepsilon_{\mathrm{mov}}(-K_X) > d-1+\epsilon$. Then*

$$\mathrm{vol}(-K_X) \leq \frac{2(d+1)^d}{\epsilon} \leq \frac{2ed^d}{\epsilon},$$

where e is Euler's number. Moreover, for any real number $C < 2e$, there exist d, ϵ, and X satisfying the assumptions such that $\mathrm{vol}(-K_X) > Cd^d/\epsilon$.

The equality for $X \cong \mathbb{P}^d$ shows that, for any fixed d, the constant $2(d+1)^d$ in Theorem 1.2 cannot be decreased, and Corollary 1.3 follows from Theorem 1.2, the inequality $(1+1/d)^d < e$, and the limit $(1+1/d)^d \to e$ as $d \to +\infty$.

Sketch of the proof. The proof of Theorem 1.2 for $d \geq 2$ has three steps.

(1) Proposition 3.1 follows [Zhu17, proof of Theorem 3.7]: X is replaced by the output $f : Y \to Z$ of a $K_{X'}$-MMP on a terminal modification X' of X, $\varepsilon_{\mathrm{mov}}(-K_X) \leq \varepsilon_{\mathrm{mov}}(-K_Y)$, and Z is a point or $\mathbb{P}^1$ by [Zhu17, Lemma 5.3]. The new point is $\mathrm{vol}(-K_X) \leq \mathrm{vol}(-K_Y)$: $H^0(X, \mathcal{O}_X(-mK_X)) = H^0(X', \mathcal{O}_{X'}(-mK_{X'}))$ as subspaces of $\mathbb{C}(X)$ for $m \geq 1$, by [Zhu17, Lemma 2.5] for the pair $(X, \frac{1}{m}(\mathrm{div}_X(u) - mK_X))$, where $u \in H^0(X, \mathcal{O}_X(-mK_X))$ is non-zero and K_X need not be $\mathbb{Q}$-Cartier.

(2) If $\varepsilon_{\mathrm{mov}}(-K_Y) > d$, then $Y \cong \mathbb{P}^d$ [Zhu17, Theorem 1.7]. If $X \ncong \mathbb{P}^d$, then $\varepsilon_{\mathrm{mov}}(-K_X) \leq d$ [Zhu17, Theorem 1.7], so by Proposition 3.1 and Lemma 6.2

$$(\varepsilon_{\mathrm{mov}}(-K_X) - d + 1)\,\mathrm{vol}(-K_X) \leq \mathrm{vol}(-K_X) \leq \mathrm{vol}(-K_Y) = (d+1)^d.$$

If $\varepsilon_{\mathrm{mov}}(-K_Y) \leq d$ and $Z \cong \mathbb{P}^1$, then Theorem 4.2 bounds $\gamma\,\mathrm{vol}(-K_Y)$ by $2d^d$, where $\gamma = \varepsilon_{\mathrm{mov}}(-K_Y) - d + 1$, using [Zhu17, Corollary 5.2], [RTZ20, Corollary 7.2], and Lemma 4.1, a canonical bundle formula for boundaries that are klt only over an open subset of $\mathbb{P}^1$.

(3) If $\varepsilon_{\mathrm{mov}}(-K_Y) \leq d$ and Z is a point, then $\rho(Y) = 1$, and Theorem 5.1 together with Lemma 6.1 bounds $\gamma\,\mathrm{vol}(-K_Y)$ by $(d+1)^d$, using [BY26, Theorem 3.2] and Proposition 5.2. Proposition 5.2 follows from [AD19, Lemma 2.17], [Liu22, Theorem 1.6], and $\rho(Y) = 1$.

The key new steps are **Steps 2 and 3** of the proof of Theorem 4.2: the divisors given by all direct summands of positive degree of $f_*\mathcal{O}_Y(-mK_Y)$ are combined into one $\mathbb{Q}$-divisor Δ with $(F_t, \Delta|_{F_t})$ lc for a general fiber $F_t \cong \mathbb{P}^{d-1}$, so that Lemma 4.1 bounds the sum of their degrees by $2m/\gamma$ times the rank.

(a) One boundary from all summands of positive degree. In [Zhu17, proof of Lemma 5.3], the canonical bundle formula is used for $(1-\lambda)B_0 + \lambda\Delta'$, where B_0 is the $\mathbb{Q}$-divisor of [Zhu17, Corollary 5.2], $\Delta' \geq 0$ satisfies $-K_Y \sim_{\mathbb{Q}} a'F + \Delta'$ for a fiber F of f, and $\lambda > 0$ depends on the singularities of Δ' along a general fiber. This yields $a'\lambda \leq 2$, but no bound for a'. In **Step 2** of the proof of Theorem 4.2, $\Delta = \frac{1}{mr_m}\sum_{a_{m,j}>0} D_j$, where $f_*\mathcal{O}_Y(-mK_Y) \cong \bigoplus_{j=1}^{r_m} \mathcal{O}_{\mathbb{P}^1}(a_{m,j})$ and

$0 \le D_j \sim -mK_Y - a_{m,j}F$ is given by the j-th summand. Then $\Delta|_{F_t} \le \frac{1}{mr_m}\sum_j \operatorname{div}(\sigma_{j,t})$ for a basis $\sigma_{1,t},\dots,\sigma_{r_m,t}$ of $H^0(F_t,\mathcal{O}_{F_t}(dm))$, and the pair $(F_t, \frac{1}{mr_m}\sum_j \operatorname{div}(\sigma_{j,t}))$ is lc by [RTZ20, Corollary 7.2], whereas $\frac{1}{m}D_j|_{F_t}$ alone can be d times a hyperplane. Thus in **Step 3**, where the rest of $-K_Y$ is taken from [Zhu17, Corollary 5.2], the coefficient of Δ can be taken close to γ, and the role of a' is played by $\alpha_m = \frac{1}{mr_m}\sum_{a_{m,j}>0} a_{m,j}$. Dividing instead by the number of summands of positive degree does not keep $(F_t, \Delta|_{F_t})$ lc, already for $d=2$, $m=1$, and the variety of Remark 4.3.

(b) An upper bound for the slopes of the tangent sheaf. Counting jets at one point only yields the lower bound $\varepsilon_{\mathrm{mov}}(-K_Y)^d \le \mathrm{vol}(-K_Y)$, so we use [BY26, Theorem 3.2], which bounds $\mathrm{vol}(-K_Y)$ from above through the slopes of T_Y. It needs $\lambda_d > 0$, where $\lambda_1 \ge \cdots \ge \lambda_d$ are the normalized slopes of Y (Definition 2.6), and its bound grows as $\lambda_1 \cdots \lambda_d$ decreases. Proposition 5.2 bounds λ_1 from above. If T_Y is not semistable, the first term $\mathcal{E}_1$ of the Harder–Narasimhan filtration of T_Y satisfies $\mu(\mathcal{E}_1) > 0$ and $\mu(\mathcal{E}_1) > \mu(\mathcal{E}_2/\mathcal{E}_1)$, so $\mathcal{E}_1$ is a foliation (Definition 2.5) by [AD19, Lemma 2.17]. Since $\rho(Y) = 1$, $-K_{\mathcal{E}_1} \equiv r\lambda_1(-K_Y)$ with $r = \operatorname{rank}\mathcal{E}_1$, and [Liu22, Theorem 1.6], which bounds the Seshadri constant of $-K_{\mathcal{E}_1}$ by r, implies $\lambda_1 \le 1/\epsilon(-K_Y) = 1/\varepsilon_{\mathrm{mov}}(-K_Y)$, where $\epsilon(-K_Y) = \varepsilon_{\mathrm{mov}}(-K_Y)$ and $((-K_Y)^d) = \mathrm{vol}(-K_Y)$ by Lemma 6.1. Since $\lambda_1 + \cdots + \lambda_d = 1$, we obtain $\lambda_1 \cdots \lambda_d \ge \gamma/\varepsilon_{\mathrm{mov}}(-K_Y)^d$, so [BY26, Theorem 3.2] loses only the factor γ^{-1}, which is compensated by the factor γ in Theorem 5.1. The lower bound $\lambda_i \ge \gamma/\varepsilon_{\mathrm{mov}}(-K_Y)$ for each i alone would lose γ^{-d}. The condition $\rho(Y) = 1$ is used only to compare $-K_{\mathcal{E}_1}$ with $-K_Y$.

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we introduce moving Seshadri constants and volumes, foliations, and the Harder–Narasimhan filtration of the tangent sheaf. In Section 3 we reduce Theorem 1.2 to Mori fiber spaces (Proposition 3.1). In Section 4 we treat fibrations over $\mathbb{P}^1$ whose general fiber is isomorphic to $\mathbb{P}^{d-1}$ (Theorem 4.2). In Section 5 we treat $\mathbb{Q}$-factorial terminal Fano varieties of Picard number one (Theorem 5.1). In Section 6 we compare the moving Seshadri constant and the volume of an ample $\mathbb{Q}$-Cartier divisor with the Seshadri constant and the self-intersection number (Lemma 6.1), and then prove Theorem 1.2 and Corollary 1.3.

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2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. Pairs.

Definition 2.1. A *pair* (X, B) consists of a normal quasi-projective variety X and an $\mathbb{R}$-divisor $B \ge 0$ on X such that $K_X + B$ is $\mathbb{R}$-Cartier. If we do not require that $B \ge 0$, then we say that (X, B) is a *sub-pair*. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \operatorname{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B).

For any (sub-)pair (X, B), we say that (X, B) is *(sub-)lc* if $a(E, X, B) \ge 0$ for any prime divisor E over X, and *(sub-)klt* if $a(E, X, B) > 0$ for any prime divisor E over X. An *lc place* of (X, B) is a prime divisor E over X such that $a(E, X, B) = 0$. An *lc center* of (X, B) is the center of an lc place of (X, B) on X.

2.2. **Moving Seshadri constants and volumes.**

Definition 2.2. Let d be a positive integer, X a normal projective variety of dimension d, and D a $\mathbb{Q}$-divisor on X. Let $\mathcal{O}_X(D) := \mathcal{O}_X(\lfloor D \rfloor)$. For any $x \in X_{\mathrm{reg}}$, we denote by $s(D,x)$ the largest non-negative integer s such that the natural map

$$H^0(X, \mathcal{O}_X(D)) \to H^0(X, \mathcal{O}_X(D) \otimes \mathcal{O}_X/\mathfrak{m}_x^{s+1})$$

is surjective, and let $s(D,x) := -1$ if no such s exists. We define

$$\varepsilon_{\mathrm{mov}}(D,x) := \limsup_{m\to+\infty} \frac{s(mD,x)}{m} \quad \text{and} \quad \varepsilon_{\mathrm{mov}}(D) := \sup_{x\in X_{\mathrm{reg}}} \varepsilon_{\mathrm{mov}}(D,x)$$

as the *moving Seshadri constant* of D at x and the *moving Seshadri constant* of D respectively. We define

$$\mathrm{vol}(D) := \limsup_{m\to+\infty} \frac{d! h^0(X, \mathcal{O}_X(mD))}{m^d}$$

as the *volume* of D.

We use the reflexive sheaves $\mathcal{O}_X(mD)$ for all positive integers m because K_X need not be $\mathbb{Q}$-Cartier, so there is no Cartier index from which to choose multiples of $-K_X$. Section 3 compares the spaces of sections in each degree m, and Lemma 6.1 in Section 6 shows that for an ample $\mathbb{Q}$-Cartier divisor, the volume and the moving Seshadri constant are computed by the self-intersection number and the Seshadri constant.

Lemma 2.3 ([Zhu17, Section 3, the paragraph before Lemma 3.1])**.** *Let d be a positive integer, X a normal projective variety of dimension d, and D a $\mathbb{Q}$-divisor on X. Then $\varepsilon_{\mathrm{mov}}(D,x) = \varepsilon_{\mathrm{mov}}(D)$ for a very general point $x \in X$. In particular, $\varepsilon_{\mathrm{mov}}(D) = \max_{x\in X_{\mathrm{reg}}} \varepsilon_{\mathrm{mov}}(D,x)$.*

Definition 2.2 is the definition in [Zhu17, formula (1) and the paragraph following it], where $\varepsilon_{\mathrm{mov}}(D)$ is defined as the maximum in Lemma 2.3, and for $\mathbb{Q}$-divisors we use the rounding convention that [Zhu17, Definition 2.14 and the following paragraph] uses for volumes. Thus the results of [Zhu17] cited in this paper hold for $\varepsilon_{\mathrm{mov}}$ and vol as in Definition 2.2.

Definition 2.4 follows [Liu22, Definition 1.4], and the Seshadri constant at a point is computed by curves as in [Laz04, Proposition 5.1.5].

Definition 2.4. Let X be a normal projective variety and D a nef $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X. For any $x \in X_{\mathrm{reg}}$, we define

$$\epsilon(D,x) := \inf_C \frac{(D\cdot C)}{\mathrm{mult}_x C},$$

where C runs over all irreducible curves on X passing through x, as the *Seshadri constant* of D at x. We define $\epsilon(D) := \sup_{x\in X_{\mathrm{reg}}} \epsilon(D,x)$.

We take the supremum over smooth points in Definition 2.4, as in [Liu22, Definition 1.4], rather than the infimum over all points, because Lemma 6.1 in Section 6 shows that the supremum is equal to the moving Seshadri constant for an ample $\mathbb{Q}$-Cartier divisor, and [Liu22, Theorem 1.6] bounds the supremum.

2.3. **Foliations and slopes.** We use the foliations of [AD19, Definitions 2.1 and 2.2].

Definition 2.5. Let X be a normal variety and $T_X := (\Omega^1_X)^*$ the tangent sheaf of X. For any torsion-free coherent sheaf $\mathcal{E} \neq 0$ on X, let $\det \mathcal{E} := (\wedge^{\mathrm{rank}\,\mathcal{E}} \mathcal{E})^{**}$. A *foliation* on X is a coherent subsheaf $0 \neq \mathcal{F} \subset T_X$ such that $T_X/\mathcal{F}$ is torsion-free and $\mathcal{F}$ is closed under the Lie bracket. For a foliation $\mathcal{F}$ on X, the *canonical class* $K_{\mathcal{F}}$ of $\mathcal{F}$ is a divisor on X such that $\mathcal{O}_X(-K_{\mathcal{F}}) \cong \det \mathcal{F}$.

We allow $\mathcal{F} = T_X$ in Definition 2.5 because Section 5 uses [AD19, Lemma 2.17] for the foliation T_X and saturated subsheaves of T_X.

Definition 2.6. Let X be a $\mathbb{Q}$-factorial normal projective variety of dimension d such that $-K_X$ is ample. For any torsion-free coherent sheaf $\mathcal{E} \neq 0$ on X, we define

$$\mu(\mathcal{E}) := \frac{c_1(\det \mathcal{E}) \cdot (-K_X)^{d-1}}{\operatorname{rank} \mathcal{E}}$$

as the *slope* of $\mathcal{E}$. We say that $\mathcal{E}$ is *semistable* if $\mu(\mathcal{E}') \leq \mu(\mathcal{E})$ for any coherent subsheaf $0 \neq \mathcal{E}' \subset \mathcal{E}$. By [GKP14, Corollary 2.27], there exists a unique filtration

$$0 = \mathcal{E}_0 \subsetneq \mathcal{E}_1 \subsetneq \cdots \subsetneq \mathcal{E}_k = T_X$$

of T_X by saturated subsheaves such that $\mathcal{E}_a/\mathcal{E}_{a-1}$ is semistable for any $1 \leq a \leq k$ and $\mu(\mathcal{E}_1/\mathcal{E}_0) > \cdots > \mu(\mathcal{E}_k/\mathcal{E}_{k-1})$. We call it the *Harder–Narasimhan filtration* of T_X. The *normalized slopes* $\lambda_1 \geq \cdots \geq \lambda_d$ of X are the real numbers

$$\frac{\mu(\mathcal{E}_a/\mathcal{E}_{a-1})}{((-K_X)^d)}, \quad 1 \leq a \leq k,$$

where each $\mu(\mathcal{E}_a/\mathcal{E}_{a-1})/((-K_X)^d)$ is repeated $\operatorname{rank}(\mathcal{E}_a/\mathcal{E}_{a-1})$ times.

The filtration in Definition 2.6 is the Harder–Narasimhan filtration of T_X in [AD19, Paragraph 2.16] with respect to the movable class $(-K_X)^{d-1}$. The slope only needs $-K_X$ to be $\mathbb{Q}$-Cartier, but we assume that X is $\mathbb{Q}$-factorial because [GKP14, Corollary 2.27], as well as [AD19, Lemma 2.17] and [Liu22, Theorem 1.6] applied to $\mathcal{E}_1$ in Section 5, concern $\mathbb{Q}$-factorial normal projective varieties, and Section 5 also needs $-K_{\mathcal{E}_1}$ to be $\mathbb{Q}$-Cartier.

3. Reduction to Mori fiber spaces

In this section, we prove Proposition 3.1, which reduces Theorem 1.2 to Mori fiber spaces over a point or over $\mathbb{P}^1$. The Mori fiber space in Proposition 3.1 is constructed as in [Zhu17, Section 5]. Under the reduction, neither factor of the product $(\varepsilon_{\mathrm{mov}}(-K_X) - d + 1)\operatorname{vol}(-K_X)$ in Theorem 1.2 decreases. Since we bound $\operatorname{vol}(-K_X)$ itself, we compare the spaces of sections in any degree as subspaces of $\mathbb{C}(X)$, and not only the moving Seshadri constants as in [Zhu17, Lemma 3.4 and Corollary 3.3]. We keep the fibration over $\mathbb{P}^1$ instead of passing to a Fano variety birational to it, since the volume is not a birational invariant (cf. [Zhu17, Example 1.3] in Section 1).

Proposition 3.1. *Let X be a normal projective variety of dimension $d \geq 2$. Assume that $\varepsilon_{\mathrm{mov}}(-K_X) > d-1$. Then there exists a Mori fiber space $f : Y \to Z$ such that Y is a $\mathbb{Q}$-factorial terminal projective variety of dimension d and the following hold.*

(1) $\operatorname{vol}(-K_X) \leq \operatorname{vol}(-K_Y)$ and $\varepsilon_{\mathrm{mov}}(-K_X) \leq \varepsilon_{\mathrm{mov}}(-K_Y)$.
(2) Either Z is a point and Y is a Fano variety with $\rho(Y) = 1$, or $Z \cong \mathbb{P}^1$ and the general fiber of f is isomorphic to $\mathbb{P}^{d-1}$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we pass to a terminal modification $h : X' \to X$ and show that $H^0(X, \mathcal{O}_X(-mK_X))$ does not change for any integer $m \geq 1$.

Let $h : X' \to X$ be a terminal modification of X, that is, h is a projective birational morphism such that X' is $\mathbb{Q}$-factorial terminal and $K_{X'}$ is h-nef [Zhu17, Definition 2.3]. Such h exists by [Zhu17, the paragraph after Definition 2.4], as a consequence of [BCHM10, Theorem 1.2(1)] applied to a resolution of X over X. We show that for any integer $m \geq 1$,

$$H^0(X', \mathcal{O}_{X'}(-mK_{X'})) = H^0(X, \mathcal{O}_X(-mK_X))$$

as subspaces of $\mathbb{C}(X) = \mathbb{C}(X')$ (here we fix a non-zero rational d-form ω and let $K_{X'} = \operatorname{div}_{X'}(\omega)$, $K_X = \operatorname{div}_X(\omega) = h_* K_{X'}$, and, in **Step 2**, $K_Y = \operatorname{div}_Y(\omega)$, so that the pushforward of $K_{X'}$ to Y is K_Y).

Let $u \in \mathbb{C}(X)$ be a non-zero rational function. If $u \in H^0(X', \mathcal{O}_{X'}(-mK_{X'}))$, then

$$\operatorname{div}_{X'}(u) - mK_{X'} \geq 0,$$

hence

$$\operatorname{div}_X(u) - mK_X = h_*\big(\operatorname{div}_{X'}(u) - mK_{X'}\big) \geq 0,$$

and $u \in H^0(X, \mathcal{O}_X(-mK_X))$.

Conversely, assume that $u \in H^0(X, \mathcal{O}_X(-mK_X))$, and let $D_X := \operatorname{div}_X(u) - mK_X \geq 0$. Although K_X need not be $\mathbb{Q}$-Cartier, $K_X + \frac{1}{m}D_X = \frac{1}{m}\operatorname{div}_X(u)$ is $\mathbb{Q}$-Cartier, so $(X, \frac{1}{m}D_X)$ is a pair. Let

$$E := \operatorname{div}_{X'}(u) - mK_{X'} - h_*^{-1}D_X.$$

Then E is a divisor, and $h_*E = \operatorname{div}_X(u) - mK_X - D_X = 0$, so E is h-exceptional. Moreover,

$$K_{X'} + \frac{1}{m}\big(h_*^{-1}D_X + E\big) = \frac{1}{m}\operatorname{div}_{X'}(u) = h^*\Big(K_X + \frac{1}{m}D_X\Big).$$

Thus, by Definition 2.1, for any prime component E_j of E (although X' need not be a log resolution, the log discrepancy of E_j can be read on X', since it can be computed on any normal model on which E_j is a divisor [KM98, Remark 2.23, Definition 2.25, and Lemma 2.30]),

$$\operatorname{mult}_{E_j} E = m\Big(1 - a\big(E_j, X, \tfrac{1}{m}D_X\big)\Big).$$

In [Zhu17, Lemma 2.5], discrepancies are used instead of log discrepancies. The discrepancy of E_j is $a(E_j, X, \frac{1}{m}D_X) - 1$. By [Zhu17, Lemma 2.5], which holds for any $\mathbb{Q}$-divisor $D \geq 0$ on X such that $K_X + D$ is $\mathbb{Q}$-Cartier and which is obtained by applying the negativity lemma [KM98, Lemma 3.39] to the h-nef divisor $K_{X'}$, we have $a(E_j, X, \frac{1}{m}D_X) \leq 1$. Hence $\operatorname{mult}_{E_j} E \geq 0$ for any j, so $E \geq 0$ and

$$\operatorname{div}_{X'}(u) - mK_{X'} = h_*^{-1}D_X + E \geq 0.$$

Therefore $u \in H^0(X', \mathcal{O}_{X'}(-mK_{X'}))$. Without the h-nefness of $K_{X'}$, sections of $\mathcal{O}_X(-mK_X)$ need not lift: if X is smooth and X'' is the blow-up of X at a point x, then X'' is $\mathbb{Q}$-factorial terminal, the exceptional divisor of $X'' \to X$ has discrepancy $d - 1 > 0$, and $H^0(X'', \mathcal{O}_{X''}(-mK_{X''}))$ consists of the sections of $\mathcal{O}_X(-mK_X)$ vanishing to order at least $m(d-1)$ at x.

Since $h^0(X', \mathcal{O}_{X'}(-mK_{X'})) = h^0(X, \mathcal{O}_X(-mK_X))$ for any integer $m \geq 1$, Definition 2.2 implies that $\operatorname{vol}(-K_{X'}) = \operatorname{vol}(-K_X)$. By [Zhu17, Lemma 3.4],

$$\varepsilon_{\mathrm{mov}}(-K_{X'}) = \varepsilon_{\mathrm{mov}}(-K_X).$$

Step 2. In this step we run a $K_{X'}$-MMP $X' \dashrightarrow Y$ ending with a Mori fiber space $f : Y \to Z$, and prove (1).

By **Step 1**, $\varepsilon_{\mathrm{mov}}(-K_{X'}) = \varepsilon_{\mathrm{mov}}(-K_X) > d - 1 > 0$, so $-K_{X'}$ is big [Zhu17, Section 3, the paragraph before Lemma 3.1]. Since X' is $\mathbb{Q}$-factorial, $K_{X'}$ is $\mathbb{Q}$-Cartier. Moreover, $K_{X'}$ is not pseudo-effective, since otherwise $0 = K_{X'} + (-K_{X'})$ would be big. Since X' is terminal, $(X', 0)$ is klt. By [BCHM10, Corollary 1.3.3], we may run a $K_{X'}$-MMP $X' \dashrightarrow Y$ which ends with a Mori fiber space $f : Y \to Z$, and $X' \dashrightarrow Y$ is a composition of divisorial contractions and flips. Since $(X', 0)$ is $\mathbb{Q}$-factorial terminal, it is dlt. By [KM98, Corollaries 3.42 and 3.43(3), Propositions 3.36 and 3.37] and induction on the number of divisorial contractions and flips, Y is a $\mathbb{Q}$-factorial terminal projective variety. Moreover, $\dim Y = d$.

Since $X' \dashrightarrow Y$ is a composition of divisorial contractions and flips, any prime divisor on Y is the strict transform of a prime divisor on X'. Let $m \geq 1$ be an integer and $u \in H^0(X', \mathcal{O}_{X'}(-mK_{X'}))$ a non-zero rational function. Then the pushforward of $\operatorname{div}_{X'}(u) - mK_{X'} \geq 0$ to Y is $\operatorname{div}_Y(u) - mK_Y$, so $\operatorname{div}_Y(u) - mK_Y \geq 0$. Thus

$$H^0(X', \mathcal{O}_{X'}(-mK_{X'})) \subset H^0(Y, \mathcal{O}_Y(-mK_Y))$$

as subspaces of $\mathbb{C}(X)$ for any integer $m \geq 1$. By Definition 2.2, $\operatorname{vol}(-K_{X'}) \leq \operatorname{vol}(-K_Y)$. By [Zhu17, Corollary 3.3], $\varepsilon_{\mathrm{mov}}(-K_{X'}) \leq \varepsilon_{\mathrm{mov}}(-K_Y)$. Together with **Step 1**, we obtain

$$\operatorname{vol}(-K_X) \leq \operatorname{vol}(-K_Y) \quad \text{and} \quad \varepsilon_{\mathrm{mov}}(-K_X) \leq \varepsilon_{\mathrm{mov}}(-K_Y),$$

which is (1).

Step 3. In this step, when $\dim Z > 0$, we verify the hypothesis $\varepsilon_{\mathrm{mov}}(-K_Y - A) \geq d-1$ of [Zhu17, Lemma 5.3] for an f-ample $\mathbb{Q}$-divisor A proportional to $-K_Y$, and deduce (2) from that lemma.

Since f is a Mori fiber space, $\rho(Y/Z) = 1$ and $-K_Y$ is f-ample. If Z is a point, then $\rho(Y) = \rho(Y/Z) = 1$ and $-K_Y$ is ample, so Y is a Fano variety with $\rho(Y) = 1$.

Assume that $\dim Z > 0$. By **Step 2**, $\varepsilon_{\mathrm{mov}}(-K_Y) \geq \varepsilon_{\mathrm{mov}}(-K_X) > d-1$. Thus there exists an integer $q \geq 2$ such that

$$\frac{q-1}{q}\varepsilon_{\mathrm{mov}}(-K_Y) \geq d-1.$$

Let $A := -\frac{1}{q}K_Y$. Then A is an f-ample $\mathbb{Q}$-divisor.

We show that for any $x \in Y_{\mathrm{reg}}$,

$$\varepsilon_{\mathrm{mov}}(-K_Y - A, x) \geq \frac{q-1}{q}\varepsilon_{\mathrm{mov}}(-K_Y, x).$$

For any Weil divisors L_1, L_2 on Y and any $x \in Y_{\mathrm{reg}}$ such that $s(L_1, x) \geq 0$ and $s(L_2, x) \geq 0$, we have

$$s(L_1 + L_2, x) \geq s(L_1, x) + s(L_2, x),$$

since sections of $\mathcal{O}_Y(L_1)$ and $\mathcal{O}_Y(L_2)$ multiply as rational functions to sections of $\mathcal{O}_Y(L_1 + L_2)$, and $\mathcal{O}_Y(L_1)$, $\mathcal{O}_Y(L_2)$, and $\mathcal{O}_Y(L_1 + L_2)$ are invertible near x [Dem92, the sentence after (6.3)]. Applying the superadditivity $q-2$ times, we obtain

$$s(-m(q-1)K_Y, x) \geq (q-1)s(-mK_Y, x)$$

for any integer $m \geq 1$ such that $s(-mK_Y, x) \geq 0$. Since $\lfloor qm(-K_Y - A) \rfloor = -m(q-1)K_Y$, we have $s(qm(-K_Y - A), x) = s(-m(q-1)K_Y, x)$ for any integer $m \geq 1$. If $\varepsilon_{\mathrm{mov}}(-K_Y, x) = 0$, then the inequality holds because $s(k(-K_Y - A), x) \geq -1$ for any integer $k \geq 1$, hence $\varepsilon_{\mathrm{mov}}(-K_Y - A, x) \geq 0$. If $\varepsilon_{\mathrm{mov}}(-K_Y, x) > 0$, then the set M_x of integers $m \geq 1$ such that $s(-mK_Y, x) \geq 0$ is infinite, and

$$\varepsilon_{\mathrm{mov}}(-K_Y, x) = \limsup_{m \in M_x, m \to +\infty} \frac{s(-mK_Y, x)}{m}.$$

Restricting to the subsequence $k = qm$ with $m \in M_x$, we obtain

$$\begin{aligned}\varepsilon_{\mathrm{mov}}(-K_Y - A, x) &\geq \limsup_{m \in M_x, m \to +\infty} \frac{s(-m(q-1)K_Y, x)}{qm} \\ &\geq \frac{q-1}{q} \limsup_{m \in M_x, m \to +\infty} \frac{s(-mK_Y, x)}{m} = \frac{q-1}{q}\varepsilon_{\mathrm{mov}}(-K_Y, x).\end{aligned}$$

Taking the supremum over $x \in Y_{\mathrm{reg}}$, we obtain

$$\varepsilon_{\mathrm{mov}}(-K_Y - A) \geq \frac{q-1}{q}\varepsilon_{\mathrm{mov}}(-K_Y) \geq d-1.$$

Since f has connected fibers, $0 < \dim Z < \dim Y = d$, and A is f-ample, by [Zhu17, Lemma 5.3], the general fiber of f is isomorphic to $\mathbb{P}^{d-1}$, and either $Z \cong \mathbb{P}^1$, or $\varepsilon_{\mathrm{mov}}(-K_Y) = d-1$ and Z is an elliptic curve. Since $\varepsilon_{\mathrm{mov}}(-K_Y) \geq \varepsilon_{\mathrm{mov}}(-K_X) > d-1$, we have $Z \cong \mathbb{P}^1$. This proves (2). □

4. Fibrations over the projective line

In this section, we prove Theorem 4.2, the case of Proposition 3.1(2) where the base of the Mori fiber space is the projective line. Apart from the normality and projectivity of X, the condition that K_X is $\mathbb{Q}$-Cartier, and the lower bound on $\varepsilon_{\mathrm{mov}}(-K_X)$, the hypotheses of Theorem 4.2 concern only the general fibers of f. Nothing is assumed about the special fibers of f or about the Cartier index of K_X, on which Proposition 3.1 provides no information. The constant $2d^d$ in Theorem 4.2 cannot be replaced by any smaller number (Remark 4.3).

Lemma 4.1 is a canonical bundle formula for boundaries that are klt only over a non-empty open subset of $\mathbb{P}^1$. In **Step 3** of the proof of Theorem 4.2, Lemma 4.1 is applied to a boundary that is not controlled over the special fibers of f.

Lemma 4.1. *Let X be a normal projective variety, $f : X \to \mathbb{P}^1$ a surjective morphism with connected fibers, and $B \geq 0$ a $\mathbb{Q}$-divisor on X such that $K_X + B$ is $\mathbb{Q}$-Cartier. Assume that*

(1) there exists a non-empty open subset $U \subset \mathbb{P}^1$ such that $(f^{-1}(U), B|_{f^{-1}(U)})$ is klt, and

(2) $K_X + B \sim_{\mathbb{Q}} f^*L$ for some $\mathbb{Q}$-divisor L on $\mathbb{P}^1$.

Then $\deg L \geq -2$.

Proof. Let $\mu : W \to X$ be a log resolution of (X, B). The sheaf $\mu_*\mathcal{O}_W(\lceil K_W - \mu^*(K_X + B)\rceil)$ is the sheaf $\mathcal{O}_X(\lceil A(X, B)\rceil)$ of [FG14, 2.3]. Here the coefficient of $K_W - \mu^*(K_X + B)$ along any prime divisor E on W is $a(E, X, B) - 1$, where $a(E, X, B)$ is the log discrepancy of Definition 2.1. Since (X, B) is klt on $f^{-1}(U)$, we have $\mathcal{O}_X(\lceil A(X, B)\rceil)|_{f^{-1}(U)} \cong \mathcal{O}_{f^{-1}(U)}$ (see the last sentence of [FG14, 2.3]). Since f has connected fibers and we are in characteristic 0, the finite part of the Stein factorization of f [Har77, Corollary III.11.5] is birational, hence an isomorphism onto the normal curve $\mathbb{P}^1$, so $f_*\mathcal{O}_X = \mathcal{O}_{\mathbb{P}^1}$. Thus $f_*\mathcal{O}_X(\lceil A(X, B)\rceil)$ has rank 1 at the general point of $\mathbb{P}^1$. By (1) and (2), $f : (X, B) \to \mathbb{P}^1$ is a klt-trivial fibration [FG14, Definition 3.1].

For any closed point $p \in \mathbb{P}^1$, let t_p be the supremum of the rational numbers t such that $(X, B + tf^*p)$ is sub-lc over a neighborhood of p. Taking a log resolution of $(X, B + f^*p)$, [KM98, Corollary 2.31(3)] implies that t_p is the minimum of finitely many rational numbers (the components of the boundary on the resolution which dominate $\mathbb{P}^1$ have coefficients < 1 by (1)), so t_p is a rational number and the supremum is attained. By generic smoothness [Har77, Corollary III.10.7], there exists a non-empty open subset $U' \subset U$ such that $f \circ \mu$ is smooth over U', and the strata of $\operatorname{Exc}(\mu) \cup \operatorname{Supp} \mu_*^{-1}B$ are smooth over U' and have no vertical components over U'. For any closed point $p \in U'$, $(f \circ \mu)^*p$ is reduced and $\operatorname{Exc}(\mu) \cup \operatorname{Supp} \mu_*^{-1}B \cup \operatorname{Supp}(f \circ \mu)^*p$ is simple normal crossing over a neighborhood of p, so [KM98, Corollary 2.31(3)] implies that $t_p = 1$. Therefore only finitely many coefficients of $\sum_p (1 - t_p)p$ are non-zero, and

$$B_{\mathbb{P}^1} := \sum_p (1 - t_p)p$$

is a $\mathbb{Q}$-divisor on $\mathbb{P}^1$, the discriminant of $f : (X, B) \to \mathbb{P}^1$ [FG14, 3.4]. Let P be any component of f^*p. Since $(X, B + t_p f^*p)$ is sub-lc over a neighborhood of p,

$$\operatorname{mult}_P B + t_p \operatorname{mult}_P f^*p \leq 1.$$

Since $B \geq 0$ and $\operatorname{mult}_P f^*p \geq 1$, we have $t_p \leq 1$. Thus $B_{\mathbb{P}^1} \geq 0$. Here we do not need (X, B) to be lc over the special fibers of f.

By [Amb04, Theorem 2.7] (see [FG14, Theorem 3.5]), the moduli part of $f : (X, B) \to \mathbb{P}^1$ is $\boldsymbol{b}$-nef. The trace of the moduli part on $\mathbb{P}^1$ is $M := L - K_{\mathbb{P}^1} - B_{\mathbb{P}^1}$ [FG14, 3.4]. Since any proper birational morphism from a normal curve to $\mathbb{P}^1$ is an isomorphism, M is nef, so $\deg M \geq 0$. Therefore

$$\deg L = \deg K_{\mathbb{P}^1} + \deg B_{\mathbb{P}^1} + \deg M \geq -2.$$

□

The proof of Theorem 4.2 combines all the direct summands of positive degree of the direct images of the pluri-anti-canonical sheaves into one boundary. In particular, it bounds the average of the positive degrees, rather than the degree of each direct summand.

Theorem 4.2. *Let X be a normal projective variety of dimension $d \geq 2$ such that K_X is $\mathbb{Q}$-Cartier, and let $f : X \to \mathbb{P}^1$ be a surjective morphism with connected fibers such that the general fiber of f is isomorphic to $\mathbb{P}^{d-1}$. Assume that $\varepsilon_{\mathrm{mov}}(-K_X) > d - 1$. Then*

$$(\varepsilon_{\mathrm{mov}}(-K_X) - d + 1) \operatorname{vol}(-K_X) \leq 2d^d.$$

Proof. Let F be the pullback under f of a closed point of $\mathbb{P}^1$. Then F is a Cartier divisor on X. For any closed point $t \in \mathbb{P}^1$, let $F_t := f^*t$. For a general closed point $t \in \mathbb{P}^1$, $F_t \cong \mathbb{P}^{d-1}$ is a smooth Cartier divisor on X. Since a local ring is regular if the quotient by a non-zero-divisor is regular, X is smooth along F_t. The smoothness of X along the general fibers of f and the condition that K_X is $\mathbb{Q}$-Cartier are needed for [Zhu17, Corollary 5.2], which we use in **Step 3**: the pair (X, D) of [Zhu17, Corollary 5.2] is defined only when $K_X + D$ is $\mathbb{Q}$-Cartier, and the criterion [Zhu17, Lemma 2.13] for klt along the general fiber of f used for [Zhu17, Corollary 5.2] concerns smooth varieties. Since $\mathcal{O}_X(F_t)|_{F_t} \cong \mathcal{O}_{F_t}$, adjunction implies that $-K_X|_{F_t} = -K_{F_t}$, that is, $\mathcal{O}_X(-K_X)|_{F_t} \cong \mathcal{O}_{\mathbb{P}^{d-1}}(d)$. Let

$$\gamma := \varepsilon_{\mathrm{mov}}(-K_X) - d + 1 > 0.$$

For any integer $k \geq 0$ and any closed point $y \in \mathbb{P}^{d-1}$, we have $s(\mathcal{O}_{\mathbb{P}^{d-1}}(k), y) = k$, since the homogeneous polynomials of degree k restrict to the polynomials of degree at most k on an affine chart containing y. Let $x \in X$ be a very general point. Then $x \in F_t$ for a general closed point $t \in \mathbb{P}^1$, and [Zhu17, Lemma 3.1] implies that

$$\varepsilon_{\mathrm{mov}}(-K_X, x) \leq \varepsilon_{\mathrm{mov}}(-K_{F_t}, x) = d.$$

By Lemma 2.3, $\varepsilon_{\mathrm{mov}}(-K_X) = \varepsilon_{\mathrm{mov}}(-K_X, x)$, so $\varepsilon_{\mathrm{mov}}(-K_X) \leq d$, that is, $\gamma \leq 1$.

The rest of the proof is long, so we divide it into several steps.

Step 1. In this step, we split $f_*\mathcal{O}_X(-mK_X)$ and construct a divisor $D_j \geq 0$ on X from each direct summand.

Fix an integer $m \geq 1$ and let $R_m := f_*\mathcal{O}_X(-mK_X)$. Since X is integral and f is surjective, R_m is torsion-free. Since the local rings of $\mathbb{P}^1$ are DVRs and any finitely generated torsion-free module over a DVR is free, R_m is locally free. Since the general fiber of f is isomorphic to $\mathbb{P}^{d-1}$ and X is smooth along each such fiber, there exists a non-empty open subset $U_1 \subset \mathbb{P}^1$ such that $f^{-1}(U_1)$ is smooth and $F_t \cong \mathbb{P}^{d-1}$ for any closed point $t \in U_1$. Thus $\mathcal{O}_X(-mK_X)|_{f^{-1}(U_1)}$ is invertible and flat over U_1 [Har77, Proposition III.9.7], and

$$h^0(F_t, \mathcal{O}_X(-mK_X)|_{F_t}) = h^0(\mathbb{P}^{d-1}, \mathcal{O}_{\mathbb{P}^{d-1}}(dm)) = \binom{dm+d-1}{d-1}$$

for any closed point $t \in U_1$. By [Har77, Corollary III.12.9], R_m has rank

$$r_m := \binom{dm+d-1}{d-1},$$

and the natural map $R_m \otimes k(t) \to H^0(F_t, \mathcal{O}_{F_t}(dm))$ is an isomorphism for any closed point $t \in U_1$. By [Har77, Exercise V.2.6],

$$R_m \cong \bigoplus_{j=1}^{r_m} \mathcal{O}_{\mathbb{P}^1}(a_{m,j})$$

for some integers $a_{m,1}, \ldots, a_{m,r_m}$. For any $1 \leq j \leq r_m$, the j-th direct summand induces a non-zero element of $H^0(\mathbb{P}^1, R_m \otimes \mathcal{O}_{\mathbb{P}^1}(-a_{m,j}))$. Since F is Cartier, the projection formula [Har77, Exercise II.5.1(d)] implies that the j-th direct summand induces a non-zero section

$$\sigma_j \in H^0(X, \mathcal{O}_X(-mK_X - a_{m,j}F)).$$

We regard σ_j as a rational function in $\mathbb{C}(X)^*$ such that $\mathrm{div}(\sigma_j) - mK_X - a_{m,j}F \geq 0$, and let

$$D_j := \mathrm{div}(\sigma_j) - mK_X - a_{m,j}F.$$

Then $D_j \geq 0$ and $D_j \sim -mK_X - a_{m,j}F$. In particular, D_j is $\mathbb{Q}$-Cartier.

Let $t \in U_1$ be a general closed point. Fixing a trivialization of $\mathcal{O}_{\mathbb{P}^1}(-a_{m,j})$ near t, we denote by $\sigma_{j,t} \in H^0(F_t, \mathcal{O}_{F_t}(dm))$ the restriction of σ_j to F_t. Since $t \in U_1$, the natural map of the

previous paragraph is an isomorphism, so $\sigma_{1,t},\dots,\sigma_{r_m,t}$ form a basis of $H^0(F_t,\mathcal{O}_{F_t}(dm))$. Since t is general, F_t is not contained in $\operatorname{Supp} D_j$, so

$$D_j|_{F_t}=\operatorname{div}(\sigma_{j,t})$$

for any $1\le j\le r_m$.

Step 2. In this step, we combine the direct summands of positive degree into a $\mathbb{Q}$-divisor Δ, and show that (X,Δ) is lc over a non-empty open subset of $\mathbb{P}^1$.

Let

$$p_m:=\frac{\#\{j\mid a_{m,j}>0\}}{r_m},\quad \alpha_m:=\frac{1}{mr_m}\sum_{a_{m,j}>0}a_{m,j},\quad\text{and}\quad \Delta:=\frac{1}{mr_m}\sum_{a_{m,j}>0}D_j.$$

Then $\Delta\ge 0$ and

$$\Delta\sim_{\mathbb{Q}}-p_mK_X-\alpha_mF.$$

For a general closed point $t\in\mathbb{P}^1$, **Step 1** implies that

$$0\le\Delta|_{F_t}\le\frac{1}{mr_m}\sum_{j=1}^{r_m}\operatorname{div}(\sigma_{j,t}).$$

By [RTZ20, Definition 2.1 and (2.2)], for the ample line bundle $-K_{F_t}\cong\mathcal{O}_{\mathbb{P}^{d-1}}(d)$ one considers the infimum of $\operatorname{lct}\big(F_t,\frac{1}{mr_m}\sum_{j=1}^{r_m}\operatorname{div}(\sigma_{j,t})\big)$ over all bases $\sigma_{1,t},\dots,\sigma_{r_m,t}$ of $H^0(F_t,\mathcal{O}_{F_t}(dm))=H^0(F_t,\mathcal{O}_{F_t}(-mK_{F_t}))$, and [RTZ20, Corollary 7.2] shows that the infimum is 1 for $\mathbb{P}^{d-1}$ and any $m\ge 1$. Hence for any basis $\sigma_{1,t},\dots,\sigma_{r_m,t}$ of $H^0(F_t,\mathcal{O}_{F_t}(dm))$, the pair $(F_t,\frac{1}{mr_m}\sum_{j=1}^{r_m}\operatorname{div}(\sigma_{j,t}))$ is lc. Since $\Delta|_{F_t}\ge 0$ and log discrepancies do not decrease when the boundary decreases, $(F_t,\Delta|_{F_t})$ is lc.

Let $\mu:W\to X$ be a log resolution of (X,Δ), and write $K_W+\Delta_W:=\mu^*(K_X+\Delta)$. By generic smoothness [Har77, Corollary III.10.7], there exists a non-empty open subset $U_2\subset U_1$ such that $f\circ\mu$ is smooth over U_2, the strata of $\operatorname{Supp}\Delta_W\cup\operatorname{Exc}(\mu)$ are smooth over U_2, and $\operatorname{Supp}\Delta_W\cup\operatorname{Exc}(\mu)$ has no vertical components over U_2. For a general closed point $t\in U_2$, let $W_t:=(f\circ\mu)^*t$. Then μ restricts to a log resolution $\mu_t:W_t\to F_t$ of $(F_t,\Delta|_{F_t})$, and adjunction on W and on X implies that

$$K_{W_t}+\Delta_W|_{W_t}=\mu_t^*(K_{F_t}+\Delta|_{F_t}).$$

Any component of $\operatorname{Supp}\Delta_W$ meeting $(f\circ\mu)^{-1}(U_2)$ dominates $\mathbb{P}^1$ and restricts to a non-zero reduced divisor on W_t, two distinct such components have no common component on W_t, since the strata of $\operatorname{Supp}\Delta_W$ are smooth over U_2, and the other components of $\operatorname{Supp}\Delta_W$ are contained in $(f\circ\mu)^{-1}(\mathbb{P}^1\backslash U_2)$, hence do not meet W_t. Thus the coefficients of $\Delta_W|_{W_t}$ are exactly the coefficients in Δ_W of the components of $\operatorname{Supp}\Delta_W$ meeting $(f\circ\mu)^{-1}(U_2)$. Since $(F_t,\Delta|_{F_t})$ is lc, the coefficients of $\Delta_W|_{W_t}$ are at most 1. By [KM98, Corollary 2.31(3)] applied to the smooth variety $(f\circ\mu)^{-1}(U_2)$, (W,Δ_W) is sub-lc over U_2. Therefore (X,Δ) is lc over U_2.

Step 3. In this step, we combine Δ and the $\mathbb{Q}$-divisor B_0 given by [Zhu17, Corollary 5.2] into a boundary B such that (X,B) is klt over a non-empty open subset of $\mathbb{P}^1$, as condition (1) of Lemma 4.1 requires (**Step 2** only shows that (X,Δ) is lc over U_2), and bound the sum of the positive degrees $a_{m,j}$ by Lemma 4.1.

Fix a rational number $0<\delta<\gamma$. Since $\gamma\le 1$, we have $1-\delta>0$. By [Zhu17, Corollary 5.2], if $g:X\to Y$ is a surjective morphism with connected fibers between normal projective varieties X and Y such that $0<\dim Y<\dim X$, the general fiber of g is isomorphic to $\mathbb{P}^{r-1}$, and $\varepsilon_{\mathrm{mov}}(-K_X)>r-\epsilon$ for some rational number $\epsilon>0$, then there exists a $\mathbb{Q}$-divisor $D\ge 0$ such that $D\sim_{\mathbb{Q}}-\frac{1}{\epsilon}K_X$ and (X,D) is klt along the general fiber of g. Since

$$\varepsilon_{\mathrm{mov}}(-K_X)=d-1+\gamma>d-(1-\delta),$$

we apply it to f with $r = d$ and $\epsilon = 1 - \delta$, and obtain a $\mathbb{Q}$-divisor $B_0 \geq 0$ such that

$$B_0 \sim_{\mathbb{Q}} -\frac{1}{1-\delta} K_X$$

and (X, B_0) is klt along the general fiber of f. Since the non-klt locus of (X, B_0) is closed and f is proper, the image of the non-klt locus of (X, B_0) is a proper closed subset of $\mathbb{P}^1$. Thus there exists a non-empty open subset $U_3 \subset U_2$ such that (X, B_0) is klt over U_3.

Let $c := 1 - (1-\delta)p_m$. Since $p_m \leq 1$, we have $c \geq \delta > 0$. Fix a rational number $0 < b < \delta/c$, and let

$$B := (1 - bp_m)(1-\delta)B_0 + b\Delta.$$

Since $c \geq \delta$, we have $b < \delta/c \leq 1$, hence $bp_m \leq b < 1$. Thus $(1 - bp_m)(1-\delta) > 0$ and $B \geq 0$. Moreover,

$$1 - (1 - bp_m)(1-\delta) - b = \delta - bc > 0.$$

Over U_3, the $\mathbb{Q}$-divisor B is the convex combination

$$B = (1 - bp_m)(1-\delta)B_0 + b\Delta + (\delta - bc) \cdot 0$$

with positive coefficients. Since log discrepancies are affine in the boundary, for any prime divisor E over X whose center on X intersects $f^{-1}(U_3)$,

$$a(E, X, B) = (1 - bp_m)(1-\delta)a(E, X, B_0) + ba(E, X, \Delta) + (\delta - bc)a(E, X, 0) > 0,$$

where $a(E, X, B_0) > 0$, $a(E, X, \Delta) \geq 0$, and $a(E, X, 0) > 0$ since $f^{-1}(U_3)$ is smooth. Thus (X, B) is klt over U_3. Moreover,

$$K_X + B \sim_{\mathbb{Q}} K_X - (1 - bp_m)K_X + b(-p_m K_X - \alpha_m F) = -b\alpha_m F.$$

In particular, $K_X + B$ is $\mathbb{Q}$-Cartier and $K_X + B \sim_{\mathbb{Q}} f^*L$ for a $\mathbb{Q}$-divisor L on $\mathbb{P}^1$ with $\deg L = -b\alpha_m$. By Lemma 4.1, $b\alpha_m \leq 2$. Letting $b \to \delta/c$, we obtain

$$\delta\alpha_m \leq 2c = 2\left(1 - (1-\delta)p_m\right).$$

Letting $\delta \to \gamma$ and multiplying by mr_m, we obtain

$$\gamma \sum_{a_{m,j}>0} a_{m,j} \leq 2\left(1 - (1-\gamma)p_m\right) mr_m \leq 2mr_m, \tag{4.1}$$

where the last inequality follows from $\gamma \leq 1$ and $p_m \geq 0$.

Step 4. In this step, we bound $\mathrm{vol}(-K_X)$.

Recall that $R_m = f_*\mathcal{O}_X(-mK_X) \cong \bigoplus_{j=1}^{r_m} \mathcal{O}_{\mathbb{P}^1}(a_{m,j})$. By [Har77, Theorem III.5.1],

$$h^0(X, \mathcal{O}_X(-mK_X)) = h^0(\mathbb{P}^1, R_m) = \sum_{j=1}^{r_m} \max(a_{m,j} + 1, 0) \leq \sum_{a_{m,j}>0} a_{m,j} + r_m.$$

By (4.1),

$$h^0(X, \mathcal{O}_X(-mK_X)) \leq \left(\frac{2}{\gamma} + \frac{1}{m}\right) mr_m.$$

Since

$$r_m = \binom{dm + d - 1}{d - 1} = \frac{d^{d-1}}{(d-1)!} m^{d-1} + O(m^{d-2}),$$

we obtain

$$\mathrm{vol}(-K_X) = \limsup_{m\to+\infty} \frac{d!h^0(X, \mathcal{O}_X(-mK_X))}{m^d} \leq \limsup_{m\to+\infty} \left(\frac{2}{\gamma} + \frac{1}{m}\right) \frac{d!r_m}{m^{d-1}} = \frac{2d^d}{\gamma}.$$

Thus $\gamma\,\mathrm{vol}(-K_X) \leq 2d^d$. □

Remark 4.3. Let $d \geq 2$ be an integer, $X := \mathbb{P}_{\mathbb{P}^1}(\mathcal{O}_{\mathbb{P}^1}^{\oplus(d-1)} \oplus \mathcal{O}_{\mathbb{P}^1}(2))$, and $f : X \to \mathbb{P}^1$ the natural morphism, with the convention that $f_*\mathcal{O}_X(1) \cong \mathcal{O}_{\mathbb{P}^1}^{\oplus(d-1)} \oplus \mathcal{O}_{\mathbb{P}^1}(2)$. Then $\varepsilon_{\mathrm{mov}}(-K_X) = d$ and $\mathrm{vol}(-K_X) = 2d^d$. Hence the constant $2d^d$ in Theorem 4.2 cannot be replaced by any smaller number.

Proof of Remark 4.3. Let

$$\mathcal{E} := \mathcal{O}_{\mathbb{P}^1}^{\oplus(d-1)} \oplus \mathcal{O}_{\mathbb{P}^1}(2),$$

and let H be a divisor on X such that $\mathcal{O}_X(H) \cong \mathcal{O}_X(1)$. By [Har77, Exercise III.8.4(b)],

$$K_X \sim -dH + f^*\left(K_{\mathbb{P}^1} + \det \mathcal{E}\right).$$

Since $\deg K_{\mathbb{P}^1} = -2$ and $\deg \det \mathcal{E} = 2$, we have $-K_X \sim dH$. Since $\mathcal{E}$ is globally generated, H is globally generated, and

$$h^0(X, \mathcal{O}_X(H)) = h^0(\mathbb{P}^1, \mathcal{E}) = d + 2.$$

Let ξ be the class of H in $A^1(X)$. By the relation defining the Chern classes [Har77, Appendix A, Section 3], $\xi^d = f^*c_1(\mathcal{E}) \cdot \xi^{d-1}$ in $A^d(X)$, since $c_i(\mathcal{E}) = 0$ for $i \geq 2$ on $\mathbb{P}^1$. Since $\deg c_1(\mathcal{E}) = 2$ and $(H^{d-1} \cdot F) = 1$ for a fiber $F \cong \mathbb{P}^{d-1}$ of f, we obtain $(H^d) = 2$.

Over $\mathbb{A}^1 := \mathbb{P}^1 \backslash \{\infty\}$ with coordinate u, we have $X|_{\mathbb{A}^1} \cong \mathbb{A}^1 \times \mathbb{P}^{d-1}$ with homogeneous coordinates $y_1, \ldots, y_{d-1}, w$ on $\mathbb{P}^{d-1}$, where w corresponds to the direct summand $\mathcal{O}_{\mathbb{P}^1}(2)$ of $\mathcal{E}$. The global sections of $\mathcal{O}_X(H)$ are spanned by $y_1, \ldots, y_{d-1}, w, uw, u^2w$. Let $y_1, \ldots, y_{d-1}, z_0, z_1, z_2$ be the corresponding homogeneous coordinates on $\mathbb{P}^{d+1}$. Thus the image of the morphism defined by $|H|$ is the quadric cone $\{z_0z_2 = z_1^2\} \subset \mathbb{P}^{d+1}$ with vertex $\{z_0 = z_1 = z_2 = 0\} \cong \mathbb{P}^{d-2}$, and the morphism is an isomorphism from $X \backslash \mathbb{P}_{\mathbb{P}^1}(\mathcal{O}_{\mathbb{P}^1}^{\oplus(d-1)})$ onto the complement of the vertex (over $\mathbb{P}^1\backslash\{0\}$, in the coordinate $v := u^{-1}$ and the homogeneous coordinate $w' := u^2w$, the sections w, uw, u^2w are v^2w', vw', w', so the roles of z_0 and z_2 are exchanged). Here $\mathbb{P}_{\mathbb{P}^1}(\mathcal{O}_{\mathbb{P}^1}^{\oplus(d-1)}) \subset X$ is the divisor given by the projection onto the direct summand $\mathcal{O}_{\mathbb{P}^1}^{\oplus(d-1)}$, and $\mathbb{P}_{\mathbb{P}^1}(\mathcal{O}_{\mathbb{P}^1}^{\oplus(d-1)}) \cap X|_{\mathbb{A}^1} = \{w = 0\}$. In particular, the morphism defined by $|H|$ is birational, and $(H^d) = 2$ is also the degree of the quadric cone.

Since $-K_X \sim dH$ is nef,

$$\mathrm{vol}(-K_X) = ((-K_X)^d) = d^d(H^d) = 2d^d$$

[Zhu17, Definition 2.14 and the paragraph after it].

Let $x \in X$ be a general point. Then $w(x) \neq 0$ and $f(x) \in \mathbb{A}^1$, so the quotients $u = uw/w$ and $y_1/w, \ldots, y_{d-1}/w$ are local coordinates near x, and each of them is a quotient of two elements of $H^0(X, \mathcal{O}_X(H))$. Thus $H^0(X, \mathcal{O}_X(H))$ generates 1-jets at x, that is, $s(H, x) \geq 1$. By the superadditivity of $s(\cdot, x)$ [Dem92, the sentence after (6.3)],

$$s(-mK_X, x) = s(mdH, x) \geq md \cdot s(H, x) \geq md$$

for any integer $m \geq 1$. Thus $\varepsilon_{\mathrm{mov}}(-K_X, x) \geq d$, so $\varepsilon_{\mathrm{mov}}(-K_X) \geq d$. Conversely, any point $y \in X$ is contained in the fiber $F_y := f^{-1}(f(y)) \cong \mathbb{P}^{d-1}$, and $\mathcal{O}_X(-K_X)|_{F_y} \cong \mathcal{O}_{\mathbb{P}^{d-1}}(d)$. By [Zhu17, Lemma 3.1], $\varepsilon_{\mathrm{mov}}(-K_X, y) \leq \varepsilon_{\mathrm{mov}}(\mathcal{O}_{\mathbb{P}^{d-1}}(d), y) = d$. Thus $\varepsilon_{\mathrm{mov}}(-K_X) = d$, and

$$(\varepsilon_{\mathrm{mov}}(-K_X) - d + 1)\,\mathrm{vol}(-K_X) = 2d^d.$$

Moreover, for this X, (4.1) is an equality for any integer $m \geq 1$. Since $-K_X \sim dH$, [Har77, Proposition II.7.11(a)] implies that

$$f_*\mathcal{O}_X(-mK_X) \cong \mathrm{Sym}^{dm} \mathcal{E},$$

which is the direct sum of the line bundles $\mathcal{O}_{\mathbb{P}^1}(2k)$, where k runs over the degrees in the last variable of the monomials of degree dm in d variables. By symmetry, the average of k over the r_m monomials of degree dm in d variables is m, so the sum of the positive degrees $a_{m,j}$ is $2mr_m$. Since $\varepsilon_{\mathrm{mov}}(-K_X) - d + 1 = 1$, both inequalities in (4.1) are equalities.

For $d = 2$ and $m = 1$, we have $f_*\mathcal{O}_X(-K_X) \cong \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(2) \oplus \mathcal{O}_{\mathbb{P}^1}(4)$, so $r_1 = 3$. Let $[y_1 : w]$ be the homogeneous coordinates on a general fiber $F_t \cong \mathbb{P}^1$ given by the direct summands $\mathcal{O}_{\mathbb{P}^1}$ and $\mathcal{O}_{\mathbb{P}^1}(2)$ of $f_*\mathcal{O}_X(1)$. The direct summands $\mathcal{O}_{\mathbb{P}^1}(2)$ and $\mathcal{O}_{\mathbb{P}^1}(4)$ of $f_*\mathcal{O}_X(-K_X)$ restrict to $y_1 w$ and w^2 on F_t, so $\Delta|_{F_t} = \frac{1}{3}[y_1 = 0] + [w = 0]$, whereas $\frac{1}{2}(\mathrm{div}(y_1 w) + \mathrm{div}(w^2)) = \frac{1}{2}[y_1 = 0] + \frac{3}{2}[w = 0]$, which has a coefficient larger than 1. Thus normalizing by the number of direct summands of positive degree instead of r_m does not lead to an lc pair on F_t. □

5. Fano varieties of Picard number one

In this section, we prove Theorem 5.1, which treats the Fano varieties of Picard number one appearing in Proposition 3.1. The volume in this case is bounded by the inequality [BY26, Theorem 3.2] once all normalized slopes are positive, and Proposition 5.2 deduces from the Seshadri condition that all normalized slopes are positive, together with a lower bound for the product of the normalized slopes.

Theorem 5.1. *Let X be a $\mathbb{Q}$-factorial terminal Fano variety of dimension $d \geq 2$ with $\rho(X) = 1$. Assume that $d - 1 < \epsilon(-K_X) \leq d$. Then*

$$(\epsilon(-K_X) - d + 1) \cdot ((-K_X)^d) \leq \left(\frac{d+1}{d}\right)^d \epsilon(-K_X)^d \leq (d+1)^d.$$

The following proposition turns the Seshadri condition into an upper bound for each normalized slope. Indeed, the first term of the Harder–Narasimhan filtration of the tangent sheaf is a foliation, and since the Picard number is one, the bound [Liu22, Theorem 1.6] for the Seshadri constant of the anti-canonical class of the foliation becomes a bound for the largest normalized slope. Together with the identity for the sum of the normalized slopes, the upper bound implies the positivity of the normalized slopes and the lower bound for their product.

Proposition 5.2. *Let X be a $\mathbb{Q}$-factorial terminal Fano variety of dimension $d \geq 2$ with $\rho(X) = 1$, and let $\lambda_1 \geq \cdots \geq \lambda_d$ be the normalized slopes of X. Assume that $d - 1 < \epsilon(-K_X) \leq d$. Then the following hold.*

(1) $\lambda_1 + \cdots + \lambda_d = 1$.
(2) $\lambda_i \leq 1/\epsilon(-K_X)$ for any $1 \leq i \leq d$.
(3) For any $1 \leq i \leq d$, we have

$$\lambda_i \geq \frac{\epsilon(-K_X) - d + 1}{\epsilon(-K_X)},$$

and

$$\lambda_1 \cdots \lambda_d \geq \frac{\epsilon(-K_X) - d + 1}{\epsilon(-K_X)^d}.$$

Proof. Let

$$0 = \mathcal{E}_0 \subsetneq \mathcal{E}_1 \subsetneq \cdots \subsetneq \mathcal{E}_k = T_X$$

be the Harder–Narasimhan filtration of T_X (Definition 2.6).

(1) Since X is normal, $\mathrm{codim}(X \setminus X_{\mathrm{reg}}) \geq 2$. The sheaves $\det T_X$ and $\mathcal{O}_X(-K_X)$ are reflexive of rank one, and they are isomorphic on X_{reg}. Thus $\det T_X \cong \mathcal{O}_X(-K_X)$ by [Har80, Proposition 1.6]. Each $\mathcal{E}_a$ and each $\mathcal{E}_a/\mathcal{E}_{a-1}$ is torsion-free, hence locally free off a closed subset of codimension at least two because X is normal. On the complement, det is multiplicative in exact sequences, and reflexive sheaves of rank one are determined off codimension two [Har80, Proposition 1.6]. Thus, as classes of $\mathbb{Q}$-Cartier divisors on the $\mathbb{Q}$-factorial variety X,

$$-K_X = c_1(\det T_X) = \sum_{a=1}^{k} c_1(\det(\mathcal{E}_a/\mathcal{E}_{a-1})).$$

Intersecting with $(-K_X)^{d-1}$, we obtain

$$((-K_X)^d) = \sum_{a=1}^{k} c_1(\det(\mathcal{E}_a/\mathcal{E}_{a-1})) \cdot (-K_X)^{d-1} = \sum_{a=1}^{k} \operatorname{rank}(\mathcal{E}_a/\mathcal{E}_{a-1})\mu(\mathcal{E}_a/\mathcal{E}_{a-1}).$$

Since $-K_X$ is ample, $((-K_X)^d) > 0$. Dividing by $((-K_X)^d)$ and using Definition 2.6, we obtain $\lambda_1 + \cdots + \lambda_d = 1$.

(2) If T_X is semistable, then $k = 1$, so $\lambda_i = 1/d$ for any i by (1). Since $\epsilon(-K_X) \leq d$, we have $\lambda_i \leq 1/\epsilon(-K_X)$.

Suppose that T_X is not semistable. Then $k \geq 2$. Let $r := \operatorname{rank} \mathcal{E}_1$. Then $1 \leq r < d$, and $\mathcal{E}_1$ is saturated in T_X. By Definition 2.6, $\mu(\mathcal{E}_1) > \mu(\mathcal{E}_2/\mathcal{E}_1)$ and $\lambda_1 > \lambda_d$. Since $\lambda_1 + \cdots + \lambda_d = 1$ by (1), we have $d\lambda_1 > 1$, that is,

$$\mu(\mathcal{E}_1) > \frac{((-K_X)^d)}{d} > 0.$$

Therefore

$$2\mu(\mathcal{E}_1) > \mu(\mathcal{E}_1) > \mu(\mathcal{E}_2/\mathcal{E}_1).$$

Since $-K_X$ is ample, $D \cdot (-K_X)^{d-1} \geq 0$ for any Cartier divisor $D \geq 0$ on X, so $(-K_X)^{d-1}$ is a movable curve class [AD19, Definition 2.15]. The slopes in Definition 2.6 are the slopes with respect to $(-K_X)^{d-1}$ in the sense of [AD19, Paragraph 2.16]. By [AD19, (2.8)], the minimal slope of $\mathcal{E}_1$ with respect to $(-K_X)^{d-1}$ is $\mu(\mathcal{E}_1)$, and the maximal slope of $T_X/\mathcal{E}_1$ with respect to $(-K_X)^{d-1}$ is $\mu(\mathcal{E}_2/\mathcal{E}_1)$, see [AD19, Paragraph 2.16]. Thus the inequality $2\mu(\mathcal{E}_1) > \mu(\mathcal{E}_2/\mathcal{E}_1)$ says that twice the minimal slope of $\mathcal{E}_1$ is larger than the maximal slope of $T_X/\mathcal{E}_1$. Since T_X is a foliation on X (Definition 2.5) and $\mathcal{E}_1$ is a saturated subsheaf of T_X, $\mathcal{E}_1$ is a foliation on X by [AD19, Lemma 2.17], which extends [CP19, Lemma 4.10] to $\mathbb{Q}$-factorial normal projective varieties.

By Definition 2.6,

$$(-K_{\mathcal{E}_1}) \cdot (-K_X)^{d-1} = c_1(\det \mathcal{E}_1) \cdot (-K_X)^{d-1} = r\mu(\mathcal{E}_1) = r\lambda_1((-K_X)^d).$$

Since $\rho(X) = 1$, comparing intersection numbers with $(-K_X)^{d-1}$ implies

$$-K_{\mathcal{E}_1} \equiv r\lambda_1(-K_X).$$

Here

$$r\lambda_1 = \frac{c_1(\det \mathcal{E}_1) \cdot (-K_X)^{d-1}}{((-K_X)^d)}$$

is a positive rational number. Since X is $\mathbb{Q}$-factorial, a positive multiple of $-K_{\mathcal{E}_1}$ is Cartier, and it is numerically equivalent to a positive rational multiple of the ample divisor $-K_X$. Thus $-K_{\mathcal{E}_1}$ is an ample $\mathbb{Q}$-Cartier divisor by Kleiman's criterion [KM98, Theorem 1.18]. In particular, $-K_{\mathcal{E}_1}$ is nef. Since $\mathcal{E}_1 \subsetneq T_X$, by [Liu22, Theorem 1.6], $\epsilon(-K_{\mathcal{E}_1})$ is at most the algebraic rank of $\mathcal{E}_1$ [Liu22, Definition 3.2], which is the rank of a subsheaf of $\mathcal{E}_1$ and hence at most r. By Definition 2.4, $\epsilon(D, x)$ depends on D only through the intersection numbers $(D \cdot C)$, so $\epsilon(D)$ depends only on the numerical class of D, and $\epsilon(tD) = t\epsilon(D)$ for any rational number $t > 0$. Thus

$$r\lambda_1\epsilon(-K_X) = \epsilon(-K_{\mathcal{E}_1}) \leq r.$$

Since $\epsilon(-K_X) > d - 1 \geq 1$, dividing by $r\epsilon(-K_X) > 0$ yields $\lambda_1 \leq 1/\epsilon(-K_X)$. Since $\lambda_i \leq \lambda_1$ for any i, (2) follows.

(3) For any $1 \leq i \leq d$, by (1) and (2), we have

$$\lambda_i = 1 - \sum_{j \neq i} \lambda_j \geq 1 - \frac{d-1}{\epsilon(-K_X)} = \frac{\epsilon(-K_X) - d + 1}{\epsilon(-K_X)}. \tag{5.1}$$

For any $1 \leq i \leq d$, let $b_i := 1 - \epsilon(-K_X)\lambda_i$. By (2) and (5.1), $b_i \in [0,1]$, and by (1),

$$\sum_{i=1}^{d} b_i = d - \epsilon(-K_X) < 1.$$

We show that $\prod_{i=1}^{\ell}(1-b_i) \geq 1 - \sum_{i=1}^{\ell} b_i$ for any $1 \leq \ell \leq d$ by induction on ℓ. For $\ell = 1$, equality holds. If the inequality holds for some $1 \leq \ell < d$, then since $1 - b_{\ell+1} \geq 0$ and $b_i \geq 0$ for any i,

$$\prod_{i=1}^{\ell+1}(1-b_i) \geq \Big(1 - \sum_{i=1}^{\ell} b_i\Big)(1-b_{\ell+1}) = 1 - \sum_{i=1}^{\ell+1} b_i + b_{\ell+1}\sum_{i=1}^{\ell} b_i \geq 1 - \sum_{i=1}^{\ell+1} b_i.$$

Taking $\ell = d$, we obtain

$$\epsilon(-K_X)^d \lambda_1 \cdots \lambda_d = \prod_{i=1}^{d}(1-b_i) \geq 1 - \sum_{i=1}^{d} b_i = \epsilon(-K_X) - d + 1,$$

which implies (3). □

To prove Theorem 5.1, we use the lower bound for the product of the normalized slopes in Proposition 5.2(3), rather than the d-th power of the lower bound for each normalized slope.

Proof of Theorem 5.1. Let $\lambda_1 \geq \cdots \geq \lambda_d$ be the normalized slopes of X. By Proposition 5.2(3),

$$\lambda_i \geq \frac{\epsilon(-K_X) - d + 1}{\epsilon(-K_X)} > 0$$

for any $1 \leq i \leq d$. Since X is terminal, X is klt. Let Y be a projective klt variety of dimension d and L an ample $\mathbb{Q}$-Cartier divisor on Y. Let $\nu_1, \ldots, \nu_d$ be the slopes of the Harder–Narasimhan filtration of T_Y with respect to L^{d-1}, divided by (L^d) and repeated according to the ranks. If $\nu_i > 0$ for any $1 \leq i \leq d$, then $(L^d) \leq ((d+1)/d)^d/(\nu_1 \cdots \nu_d)$ by [BY26, Theorem 3.2]. We apply it to $Y = X$ and $L = -K_X$. For $L = -K_X$, the numbers $\nu_1, \ldots, \nu_d$ are the normalized slopes $\lambda_1, \ldots, \lambda_d$ of Definition 2.6, and X need not be Gorenstein [BY26, Remark 3.3]. Hence [BY26, Theorem 3.2] implies

$$((-K_X)^d) \leq \left(\frac{d+1}{d}\right)^d \frac{1}{\lambda_1 \cdots \lambda_d}.$$

Multiplying by $\epsilon(-K_X) - d + 1 > 0$ and using Proposition 5.2(3), we obtain

$$(\epsilon(-K_X) - d + 1) \cdot ((-K_X)^d) \leq \left(\frac{d+1}{d}\right)^d \frac{\epsilon(-K_X) - d + 1}{\lambda_1 \cdots \lambda_d} \leq \left(\frac{d+1}{d}\right)^d \epsilon(-K_X)^d.$$

Since $\epsilon(-K_X) \leq d$, we have $\left(\frac{d+1}{d}\right)^d \epsilon(-K_X)^d \leq (d+1)^d$. □

Remark 5.3. Under the conditions $\lambda_i \leq 1/\epsilon(-K_X)$ for any i and $\lambda_1 + \cdots + \lambda_d = 1$, the lower bound for $\lambda_1 \cdots \lambda_d$ in Proposition 5.2(3) cannot be improved. Indeed, $\log(\lambda_1 \cdots \lambda_d)$ is a concave function on the polytope defined by $\lambda_i \leq 1/\epsilon(-K_X)$ and $\lambda_1 + \cdots + \lambda_d = 1$ (on the polytope $\lambda_i = 1 - \sum_{j \neq i} \lambda_j \geq (\epsilon(-K_X) - d + 1)/\epsilon(-K_X) > 0$, so it is compact and the logarithm is defined), so the minimum is attained at a vertex, that is, at a point where $d-1$ coordinates are equal to $1/\epsilon(-K_X)$ and the other coordinate is equal to $(\epsilon(-K_X) - d + 1)/\epsilon(-K_X)$.

6. Proofs of the main theorems

In this section, we prove Theorem 1.2 and Corollary 1.3.

We first prove Lemma 6.1, which compares $\varepsilon_{\mathrm{mov}}$ and vol with the Seshadri constant and the intersection number used in Theorem 5.1. After Lemma 6.1 and the computation for $\mathbb{P}^d$ in Lemma 6.2, we combine Proposition 3.1 with Theorems 4.2 and 5.1.

Lemma 6.1. *Let d be a positive integer, X a normal projective variety of dimension d, and D an ample $\mathbb{Q}$-Cartier divisor on X. Then $\mathrm{vol}(D) = (D^d)$, and $\varepsilon_{\mathrm{mov}}(D, x) = \epsilon(D, x)$ for any $x \in X_{\mathrm{reg}}$. In particular, $\varepsilon_{\mathrm{mov}}(D) = \epsilon(D)$.*

Proof. Let r be a positive integer such that rD is Cartier, and let $\mathcal{L} := \mathcal{O}_X(rD)$. Then $\mathcal{L}$ is an ample invertible sheaf, and $\mathcal{O}_X(krD) = \mathcal{L}^k$ for any positive integer k.

We first prove the lower bounds. Since $\mathrm{vol}(D)$ and $\varepsilon_{\mathrm{mov}}(D, x)$ are defined by lim sup over all positive integers m, we may take the lim sup along the subsequence $m = kr$. By [Laz04, Theorem 1.1.24] applied to rD, we have $\chi(X, \mathcal{L}^k) = ((rD)^d)k^d/d! + O(k^{d-1})$. Since $\mathcal{L}$ is ample, $H^i(X, \mathcal{L}^k) = 0$ for any $i > 0$ and any $k \gg 0$ [Har77, Proposition III.5.3]. Since $((rD)^d) = r^d(D^d)$, we have

$$h^0(X, \mathcal{L}^k) = \frac{r^d(D^d)}{d!}k^d + O(k^{d-1}),$$

hence

$$\lim_{k\to+\infty} \frac{d!h^0(X, \mathcal{O}_X(krD))}{(kr)^d} = (D^d) \quad \text{and} \quad \mathrm{vol}(D) \geq (D^d). \tag{6.1}$$

Fix $x \in X_{\mathrm{reg}}$. By [Laz04, Theorem 5.1.17] (cf. [Dem92, Theorem 6.4]), which holds for an ample divisor on any irreducible projective variety at a smooth point, and by [Laz04, Proposition 5.1.5], which shows that the Seshadri constant defined there by blowing up x coincides with the one in Definition 2.4, we have

$$\lim_{k\to+\infty} \frac{s(krD, x)}{k} = \epsilon(rD, x) = r\epsilon(D, x),$$

where the last equality follows from Definition 2.4, since $(rD \cdot C) = r(D \cdot C)$ for any curve C through x. Thus

$$\lim_{k\to+\infty} \frac{s(krD, x)}{kr} = \epsilon(D, x) \quad \text{and} \quad \varepsilon_{\mathrm{mov}}(D, x) \geq \epsilon(D, x). \tag{6.2}$$

We now prove the upper bounds. Fix $x \in X_{\mathrm{reg}}$. Since $\mathcal{L}$ is ample, there exists a positive integer a such that $\mathcal{O}_X(-jD) \otimes \mathcal{L}^a$ is globally generated for any integer $0 \leq j < r$. Since $(ar-j)D = -jD+a(rD)$ as divisors and rD is Cartier, we have $\mathcal{O}_X((ar-j)D) \cong \mathcal{O}_X(-jD)\otimes\mathcal{L}^a$. Since X is smooth at x, $\mathcal{O}_X((ar-j)D)$ is invertible near x. Thus for any integer $0 \leq j < r$, there exists $\sigma_j \in H^0(X, \mathcal{O}_X((ar-j)D))$ which does not vanish at x.

Let m be a positive integer, and write $m = qr + j$ with integers $q \geq 0$ and $0 \leq j < r$. Since $mD + (ar-j)D = (q+a)rD$ as divisors, multiplication by σ_j defines an injective homomorphism

$$\mathcal{O}_X(mD) \to \mathcal{O}_X((q+a)rD) = \mathcal{L}^{q+a},$$

which is an isomorphism over an open neighborhood of x, because σ_j is a unit near x. Thus $h^0(X, \mathcal{O}_X(mD)) \leq h^0(X, \mathcal{L}^{q+a})$. Moreover, for any non-negative integer s, multiplication by σ_j induces an isomorphism

$$H^0(X, \mathcal{O}_X(mD) \otimes \mathcal{O}_X/\mathfrak{m}_x^{s+1}) \to H^0(X, \mathcal{L}^{q+a} \otimes \mathcal{O}_X/\mathfrak{m}_x^{s+1})$$

compatible with the restriction maps from $H^0(X, \mathcal{O}_X(mD))$ and $H^0(X, \mathcal{L}^{q+a})$. Hence, if the restriction map from $H^0(X, \mathcal{O}_X(mD))$ to the left-hand side of the isomorphism induced by σ_j is surjective, then so is the restriction map from $H^0(X, \mathcal{L}^{q+a})$ to the right-hand side. Thus $s(mD, x) \leq s((q+a)rD, x)$. Therefore,

$$\frac{d!h^0(X, \mathcal{O}_X(mD))}{m^d} \leq \frac{d!h^0(X, \mathcal{O}_X((q+a)rD))}{((q+a)r)^d} \cdot \left(\frac{(q+a)r}{m}\right)^d$$

and

$$\frac{s(mD, x)}{m} \leq \frac{s((q+a)rD, x)}{(q+a)r} \cdot \frac{(q+a)r}{m}.$$

When $m \to +\infty$, we have $q \to +\infty$ and $(q+a)r/m \to 1$. By (6.1) and (6.2), the right-hand sides converge to (D^d) and $\epsilon(D, x)$ respectively. Hence $\mathrm{vol}(D) \leq (D^d)$ and $\varepsilon_{\mathrm{mov}}(D, x) \leq \epsilon(D, x)$.

Thus $\mathrm{vol}(D) = (D^d)$, and $\varepsilon_{\mathrm{mov}}(D, x) = \epsilon(D, x)$ for any $x \in X_{\mathrm{reg}}$. Taking the supremum over X_{reg} on both sides, we obtain $\varepsilon_{\mathrm{mov}}(D) = \epsilon(D)$. □

Lemma 6.2. *Let d be a positive integer. Then $\varepsilon_{\mathrm{mov}}(-K_{\mathbb{P}^d}) = d+1$ and $\mathrm{vol}(-K_{\mathbb{P}^d}) = (d+1)^d$.*

Proof. Let H be a hyperplane in $\mathbb{P}^d$. Then $-K_{\mathbb{P}^d} \sim (d+1)H$ [Har77, Example II.8.20.1], which is ample. By Lemma 6.1 and [Laz04, Examples 5.1.3, 5.1.4, and 5.1.7], we have $\varepsilon_{\mathrm{mov}}(-K_{\mathbb{P}^d}) = \epsilon(-K_{\mathbb{P}^d}) = (d+1)\epsilon(H) = d+1$ and $\mathrm{vol}(-K_{\mathbb{P}^d}) = (((d+1)H)^d) = (d+1)^d$. □

Proof of Theorem 1.2. (a) The case $X \cong \mathbb{P}^d$. By Lemma 6.2,

$$(\varepsilon_{\mathrm{mov}}(-K_{\mathbb{P}^d}) - d + 1)\,\mathrm{vol}(-K_{\mathbb{P}^d}) = 2(d+1)^d,$$

so the equality holds if $X \cong \mathbb{P}^d$.
(b) The case $d = 1$. By [Har77, Theorem I.6.2A], X is a smooth projective curve. Suppose that $\deg K_X \geq 0$. Then $\deg(-mK_X) \leq 0$ for any positive integer m, so $h^0(X, \mathcal{O}_X(-mK_X)) \leq 1$ by [Har77, Lemma IV.1.2]. Since $\dim_{\mathbb{C}} \mathcal{O}_X/\mathfrak{m}_x^2 = 2$ for any $x \in X$, we have $s(-mK_X, x) \leq 0$. Hence $\varepsilon_{\mathrm{mov}}(-K_X) \leq 0$, which contradicts $\varepsilon_{\mathrm{mov}}(-K_X) > d - 1 = 0$. Thus $\deg K_X < 0$. Let g be the genus of X. Since $\deg K_X = 2g - 2$ [Har77, Example IV.1.3.3], we have $g = 0$, hence $X \cong \mathbb{P}^1$ [Har77, Example IV.1.3.5]. By (a), $(\varepsilon_{\mathrm{mov}}(-K_X) - d + 1)\,\mathrm{vol}(-K_X) = 2(d+1)^d$.
(c) The case $d \geq 2$. By Proposition 3.1, there exists a Mori fiber space $f : Y \to Z$ such that Y is a $\mathbb{Q}$-factorial terminal projective variety of dimension d, $\mathrm{vol}(-K_X) \leq \mathrm{vol}(-K_Y)$, $\varepsilon_{\mathrm{mov}}(-K_X) \leq \varepsilon_{\mathrm{mov}}(-K_Y)$, and either Z is a point and Y is a Fano variety with $\rho(Y) = 1$, or $Z \cong \mathbb{P}^1$ and the general fiber of f is isomorphic to $\mathbb{P}^{d-1}$. Since $\varepsilon_{\mathrm{mov}}(-K_X) - d + 1 > 0$ and $\mathrm{vol}(-K_Y) \geq 0$, we have

$$\begin{aligned}(\varepsilon_{\mathrm{mov}}(-K_X) - d + 1)\,\mathrm{vol}(-K_X) &\leq (\varepsilon_{\mathrm{mov}}(-K_X) - d + 1)\,\mathrm{vol}(-K_Y)\\ &\leq (\varepsilon_{\mathrm{mov}}(-K_Y) - d + 1)\,\mathrm{vol}(-K_Y).\end{aligned} \tag{6.3}$$

In particular, $\varepsilon_{\mathrm{mov}}(-K_Y) > d - 1$. By (6.3), it suffices to show that

$$(\varepsilon_{\mathrm{mov}}(-K_Y) - d + 1)\,\mathrm{vol}(-K_Y) \leq 2(d+1)^d.$$

We consider three cases.

Case 1. $\varepsilon_{\mathrm{mov}}(-K_Y) > d$. By [Zhu17, Theorem 1.7], $Y \cong \mathbb{P}^d$. By (a),

$$(\varepsilon_{\mathrm{mov}}(-K_Y) - d + 1)\,\mathrm{vol}(-K_Y) = 2(d+1)^d.$$

Case 2. $\varepsilon_{\mathrm{mov}}(-K_Y) \leq d$ and $Z \cong \mathbb{P}^1$. Since Y is terminal, K_Y is $\mathbb{Q}$-Cartier. Since f is a Mori fiber space, f is a surjective morphism with connected fibers. By Theorem 4.2 applied to f,

$$(\varepsilon_{\mathrm{mov}}(-K_Y) - d + 1)\,\mathrm{vol}(-K_Y) \leq 2d^d < 2(d+1)^d.$$

Case 3. $\varepsilon_{\mathrm{mov}}(-K_Y) \leq d$ and Z is a point. Then Y is a $\mathbb{Q}$-factorial terminal Fano variety with $\rho(Y) = 1$, and $-K_Y$ is an ample $\mathbb{Q}$-Cartier divisor. By Lemma 6.1, $\epsilon(-K_Y) = \varepsilon_{\mathrm{mov}}(-K_Y)$ and $((-K_Y)^d) = \mathrm{vol}(-K_Y)$. In particular, $d - 1 < \epsilon(-K_Y) \leq d$. By Theorem 5.1,

$$(\varepsilon_{\mathrm{mov}}(-K_Y) - d + 1)\,\mathrm{vol}(-K_Y) = (\epsilon(-K_Y) - d + 1) \cdot ((-K_Y)^d) \leq (d+1)^d < 2(d+1)^d.$$

By Proposition 3.1, Cases 1–3 exhaust all possibilities. Thus, by (6.3),

$$(\varepsilon_{\mathrm{mov}}(-K_X) - d + 1)\,\mathrm{vol}(-K_X) \leq 2(d+1)^d.$$

□

Proof of Corollary 1.3. Since $\varepsilon_{\mathrm{mov}}(-K_X) > d - 1 + \epsilon$, we have $\varepsilon_{\mathrm{mov}}(-K_X) - d + 1 > \epsilon > 0$. By Theorem 1.2,

$$\mathrm{vol}(-K_X) \leq \frac{2(d+1)^d}{\varepsilon_{\mathrm{mov}}(-K_X) - d + 1} \leq \frac{2(d+1)^d}{\epsilon}.$$

Since $\log(1+1/d) < 1/d$, we have $(1+1/d)^d < e$, hence

$$2(d+1)^d = 2\left(1+\frac{1}{d}\right)^d d^d < 2ed^d.$$

This proves $\mathrm{vol}(-K_X) \leq 2(d+1)^d/\epsilon \leq 2ed^d/\epsilon$.

Now let $C < 2e$ be a real number. Since $(1+1/d)^d \to e$ when $d \to +\infty$, there exists a positive integer d such that $2(1+1/d)^d > C$. Then $C(1+1/d)^{-d} < 2$, so we may pick a real number ϵ such that

$$\max\left\{0, C\left(1+\frac{1}{d}\right)^{-d}\right\} < \epsilon < 2.$$

Let $X := \mathbb{P}^d$. By Lemma 6.2, $\varepsilon_{\mathrm{mov}}(-K_X) = d+1$ and $\mathrm{vol}(-K_X) = (d+1)^d$. Since $\epsilon < 2$, we have $\varepsilon_{\mathrm{mov}}(-K_X) = d+1 > d-1+\epsilon$, so d, ϵ, and X satisfy the assumptions. Since $(1+1/d)^d\epsilon > C$ and $\epsilon > 0$, we have

$$\mathrm{vol}(-K_X) = \left(1+\frac{1}{d}\right)^d d^d > \frac{Cd^d}{\epsilon}.$$

□

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School of Mathematical Sciences, Peking University, No. 5 Yiheyuan Road, Haidian District, Beijing 100871, China

Beijing International Center for Mathematical Research, Peking University, No. 5 Yiheyuan Road, Haidian District, Beijing 100871, China

Email address: liujihao@math.pku.edu.cn